AE 153 – Agricultural Meteorology & Climatology (2,1,2)

Lecturer: E. Bessah

Course Content

- Importance and practical applications of agro-meteorology measurements
- Climatological data processing and analysis
- Climate variability and change

Course Objectives

Specific Objectives:

1. To measure agro-meteorology data
2. To determine the practical applications of the agro-meteorology data
3. To process and analyze weather/climate data
4. To explain solar radiation relationship with the atmosphere
5. To explain climate variability and climate change

Reference Material

1. Soil and water conservation engineering by G. O. Schwab
2. Hydrology in practice by E. M. Shaw
3. Agrometeorology: principles and applications of climate studies in agriculture by H S Mavi and G. J. Tupper

Teminologies

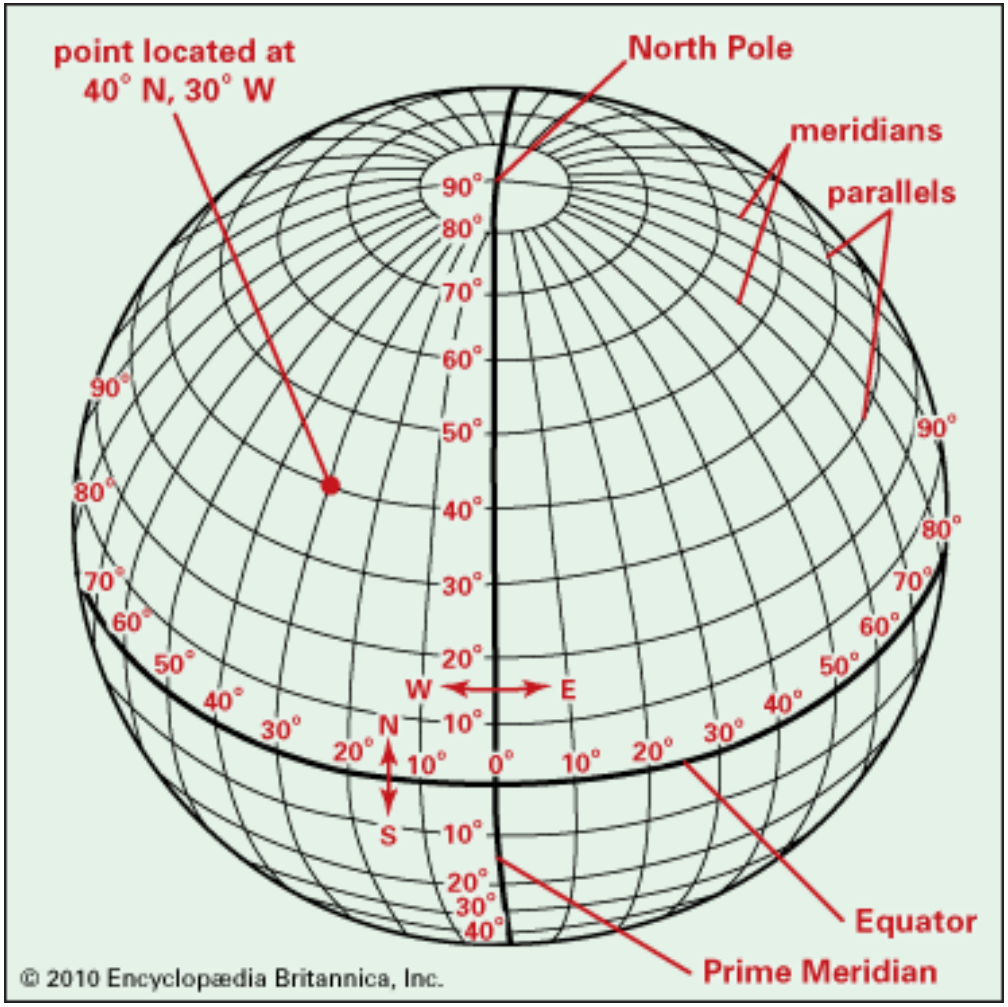
Weather: The **state of the atmosphere at a given time and place**, with respect to variables such as **rainfall, temperature, moisture, wind velocity, and atmospheric pressure**.

That is the present condition of these variables or elements and their variations over periods of up to say two weeks.

Climate: Is the **statistics of variables** (such as temperature, humidity, atmospheric pressure, wind, rainfall and other meteorological elements in a given region over long period of time (usually 30 years).

Factors that determine/influences the climate of a location:

- Latitude
- Terrain
- Vegetation
- Altitude
- Presence of water bodies and their currents



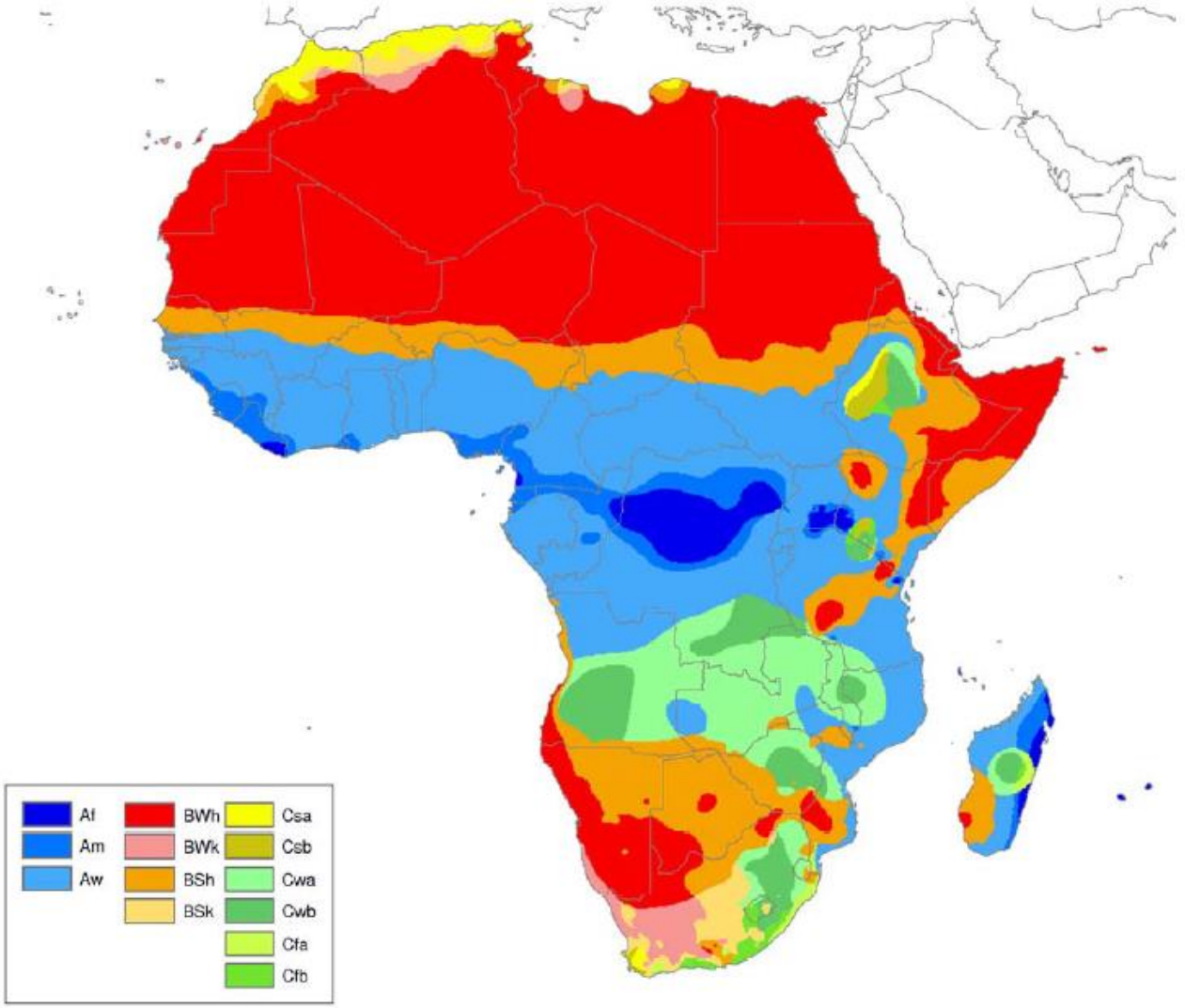
Source: Britannica

Climate classification

Is done using the average climatic conditions and the typical ranges of different variables, most commonly using temperature and precipitation.

The most commonly used classification scheme was originally developed by Köppen and Thornthwaite.

- Köppen classification: The five main groups are: A (Tropical); B (Dry); C (Temperate); D (Continental) and E (Polar)



Source: Peel et al., 2007

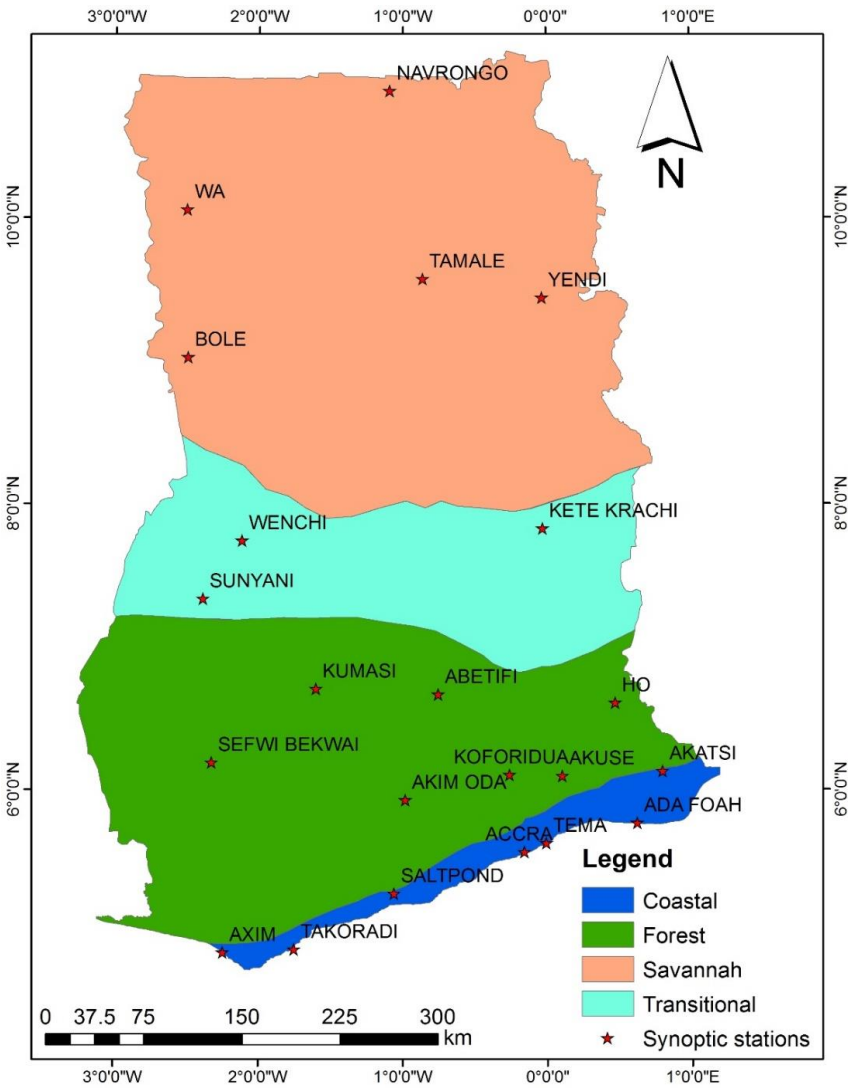
Table 1. Description of Köppen climate symbols and defining criteria.

1st	2nd	3rd	Description	Criteria*
A			Tropical	$T_{cold} \geq 18$
	f		- Rainforest	$P_{dry} \geq 60$
	m		- Monsoon	Not (Af) & $P_{dry} \geq 100 - MAP/25$
	w		- Savannah	Not (Af) & $P_{dry} < 100 - MAP/25$
B			Arid	$MAP < 10 \times P_{threshold}$
	W		- Desert	$MAP < 5 \times P_{threshold}$
	S		- Steppe	$MAP \geq 5 \times P_{threshold}$
		h	- Hot	$MAT \geq 18$
		k	- Cold	$MAT < 18$
C			Temperate	$T_{hot} > 10$ & $0 < T_{cold} < 18$
	s		- Dry Summer	$P_{sdry} < 40$ & $P_{sdry} < P_{wwet}/3$
	w		- Dry Winter	$P_{wdry} < P_{swet}/10$
	f		- Without dry season	Not (Cs) or (Cw)
		a	- Hot Summer	$T_{hot} \geq 22$
		b	- Warm Summer	Not (a) & $T_{mon10} \geq 4$
		c	- Cold Summer	Not (a or b) & $1 \leq T_{mon10} < 4$
D			Cold	$T_{hot} > 10$ & $T_{cold} \leq 0$
	s		- Dry Summer	$P_{sdry} < 40$ & $P_{sdry} < P_{wwet}/3$
	w		- Dry Winter	$P_{wdry} < P_{swet}/10$
	f		- Without dry season	Not (Ds) or (Dw)
		a	- Hot Summer	$T_{hot} \geq 22$
		b	- Warm Summer	Not (a) & $T_{mon10} \geq 4$
		c	- Cold Summer	Not (a, b or d)
		d	- Very Cold Winter	Not (a or b) & $T_{cold} < -38$
E			Polar	$T_{hot} < 10$
	T		- Tundra	$T_{hot} > 0$
	F		- Frost	$T_{hot} \leq 0$

*MAP = mean annual precipitation, MAT = mean annual temperature, T_{hot} = temperature of the hottest month, T_{cold} = temperature of the coldest month, T_{mon10} = number of months where the temperature is above 10, P_{dry} = precipitation of the driest month, P_{sdry} = precipitation of the driest month in summer, P_{wdry} = precipitation of the driest month in winter, P_{swet} = precipitation of the wettest month in summer, P_{wwet} = precipitation of the wettest month in winter, $P_{threshold}$ = varies according to the following rules (if 70% of MAP occurs in winter then $P_{threshold}$ = 2 x MAT, if 70% of MAP occurs in summer then $P_{threshold}$ = 2 x MAT + 28, otherwise $P_{threshold}$ = 2 x MAT + 14). Summer (winter) is defined as the warmer (cooler) six month period of ONDJFM and AMJJAS.

Source: Peel et al., 2007

Compiled by NKay Study



Source: Tanu et al….

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Teminologies

Meteorology: The **scientific study of** the **atmosphere** and **atmospheric conditions**, especially as they relate to weather and weather forecasting.

Agro-meteorology: The branch of meteorology that studies the **effect of weather and climate on agriculture**. It examines the effects and impacts of weather and climate on crops, rangeland, livestock, and various agricultural operations.

Agricultural meteorologists: **collect, analyse and interpret weather and climate data** needed to understand the interactions between vegetation and animals and their atmospheric environments.

Benefits of agro-meteorological information

Agro-meteorologists:

1. provide information valuable in making proper decisions
   a. managing resources consumed by agriculture
   b. optimizing agricultural production
   c. adopting farming practices that minimizes adverse effects of agriculture on the environment
2. provide information ensuring economic and environmental sustainability of agric. now and in the future
3. quantify, evaluate and provide information on the impact and consequences of climate variability and change on agriculture
4. assist policy makers in developing strategies to deal with extreme climatic events such as floods, rain storms, or droughts and climatic changes such as global warming and climate variability

Compiled by NKay Study

AE 153 – Agricultural Meteorology & Climatology (2,1,2)

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AGRO-METEOROLOGY MEASUREMENTS

- Weather occurs in the part of the earth's atmosphere called the troposphere.
- The troposphere is the layer which is closest to the earth's surface, or the sky that we can see.
- The weather is the condition of the atmosphere (or changes in the atmosphere) at a particular time and place.

AGRO-METEOROLOGY MEASUREMENTS

- The weather is shaped by 4 main factors:

  1. Air
  2. Sun
  3. Water
  4. Earth's movement

- Weather instruments focuses on the primary task of measuring atmospheric conditions.

- Meteorologists are interested in the parameters that influences different types of weather, including:

  pressure, heat, moisture and wind.

AGRO-METEOROLOGY MEASUREMENTS

Key weather variables, units and measuring equipment used

Weather variable	Units	Equipment
Rainfall	mm	Standard Rain gauge, Automatic Rain gauge (continuous record of rainfall)
Wind run/speed	m/s	Cup Anemometer
Wind direction	°	wind vane
Sunshine	hour	Stoke sunshine recorder
Evaporation	mm	Evaporation Pan, Piche Evaporimeter
Dry & Wet bulb temperature	°C	Dry & Wet bulb Thermometer
Min & Max temperature	°C	Minimum & Maximum Thermometer
Temp & Humidity	°C & %	Thermohygrograph
Soil temperature	°C	Soil Thermometer
Atmospheric pressure	mbar	Barometer
Solar-radiation	W/m^2	Radiometer
Relative humidity	%	Psychrometer, hygrometer

Weather station on a farm



AGRO-METEOROLOGY MEASUREMENTS

Stevenson screen

- It is a meteorological screen (white louvered box) used to house some weather instruments to shield them from precipitation and direct heat radiation from outside sources while still allowing air to circulate freely around them.
- Stevenson screen houses four types of thermometers (maximum, minimum, dry bulb and wet bulb-temperature), hygrometer, psychrometer, barometer and thermograph.
- It is usually placed at about 2 m height from the ground.
- Have two doors one facing the north and the other south

AGRO-METEOROLOGY MEASUREMENTS

Stevenson screen



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Stevenson screen

Location of a Stevenson Screen

The stevenson screen is a small weather station that is used to measure local weather conditions. A weather centre will include a thermometer (max/min) a hygrometer (wet/dry bulb thermometer) and a barometer. Outside the Stevenson screen there will be rain gauge, an anemometer and a wind vane.

Height Above ground: The stevenson screen should be above the ground (at least 1.2 metres) so that it does not receive heat released from the ground.

The screen has a roof so to protect the instruments from precipitation. Rain could affect temperature or humidity readings.

Away from buildings: To avoid any heat from buildings, or any shadows from buildings, the Stevenson screen should be placed away from them.



The door of the screen should always face away from the sun. This means that in the northern hemisphere it should face north.

The screen is painted white so as to reflect incoming solar radiation. It does not want to absorb heat because if it does, it will not give an accurate reading of the air.

The screen has slats to allow air to circulate freely around the instruments.

In the open: The Stevenson screen should not only be away from buildings, but also trees, hills, etc. This will ensure that air is allowed to circulate and that the screen is not in the shade. The screen should be twice the distance away from an object as the objects height e.g. if a tree is 10 metres, then it should be 20 metres away.

Grass surface (albedo): To avoid reflection from white surfaces or absorption from dark surfaces, Stevenson screens are usually placed on grass. Grass has an albedo of 23%.

Stevenson screen



Stevenson screen



AGRO-METEOROLOGY MEASUREMENTS

Precipitation

Precipitation types: Precipitation may be in the form of

1. Rain: Falling water droplet
2. Drizzle: Uniform water with droplet sizes less that 0.5 mm
3. Snow: Frozen water particles
4. Sleet: Raindrops fall through below freezing temperature air
5. Hail: Solid precipitation made up of balls or irregular lumps of ice
6. Dew: formed from the direct condensation and absorption from the atmosphere.

AGRO-METEOROLOGY MEASUREMENTS

Precipitation

Formation of Precipitation: There are three primary steps for the formation of precipitation -

A. Creation of saturation conditions in the atmosphere (typically when there is lifting of air mass)

B. Phase change from vapour to liquid (cloud formation)

C. Growth of droplets to appreciable size that will be able to overcome upward velocity

AGRO-METEOROLOGY MEASUREMENTS

A. Creation of Saturation Conditions

1. **Air Mass movement**: The characteristics of the air mass are based on where it originated and is subsequently so named.

   For example: Tropical continental (Tc) (hot and dry wind from deserts in low latitude e.g. Sahara) and Tropical maritime (Tm) air mass (warm and moist from sub-tropical oceans)

AGRO-METEOROLOGY MEASUREMENTS

A. Creation of Saturation Conditions

2. Lifting Mechanisms

Frontal systems:

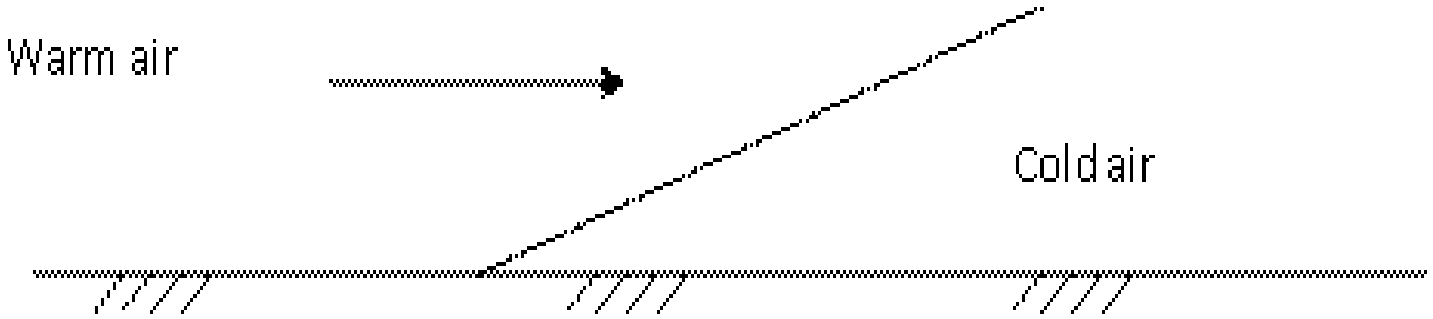
- The boundary between two adjacent air masses of different temperature and moisture content.
- Boundary between air masses with different properties is called a *front*

AGRO-METEOROLOGY MEASUREMENTS

A. Creation of Saturation Conditions

2. Lifting Mechanisms

i. **Warm Front**: Warm air advancing on colder air or when warm air overrides cold air

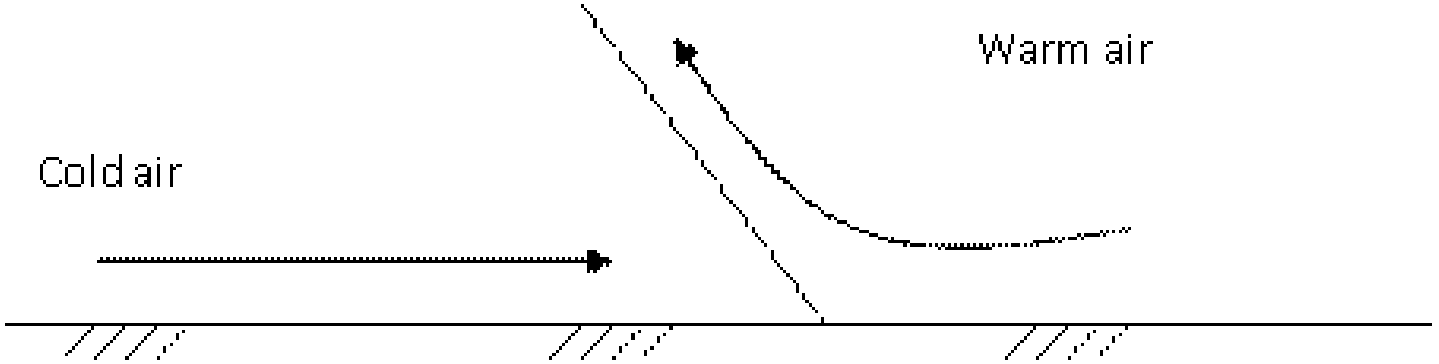


The convective rise (slope) of warm air is mild, with associated precipitation events having longer duration, lower intensity, and less energy resulting in gentler storms.
Durations: 3 – 10 h; Intensities: 1.25 – 5 mm/h

AGRO-METEOROLOGY MEASUREMENTS

A. Creation of Saturation Conditions

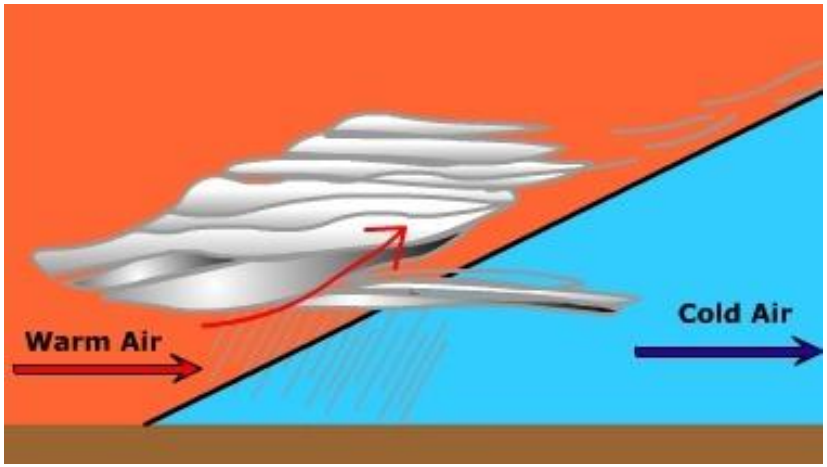
ii. **Cold front**: Cold air advancing on warm air



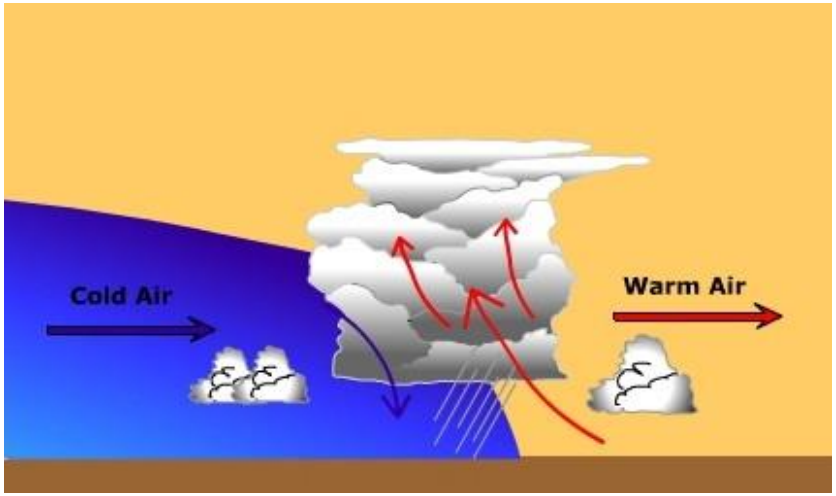
The convective rise (slope) is much steeper and can be nearly vertical. The associated precipitation events tend to have shorter duration, higher intensity, and more energy resulting in stronger more severe storms (the air movement is much faster).

Durations: 2 – 4 h; Intensities: 2.5 – 13.0 mm/h

Frontal Lifting



Warm front (produces stratus cloud)



Cold front (produces cumulus cloud)

Frontal Lifting



Status cloud “characterized by horizontal layering with a uniform base”

Frontal Lifting



Cumulus clouds “from latin word cumulo-, meaning heap or pile”

AGRO-METEOROLOGY MEASUREMENTS

A. Creation of Saturation Conditions

3. Cyclonic Events

i. *Tropical Cyclones*: These are non-frontal events but convective and generated by rising of warm-moist air masses. Typical requirements:

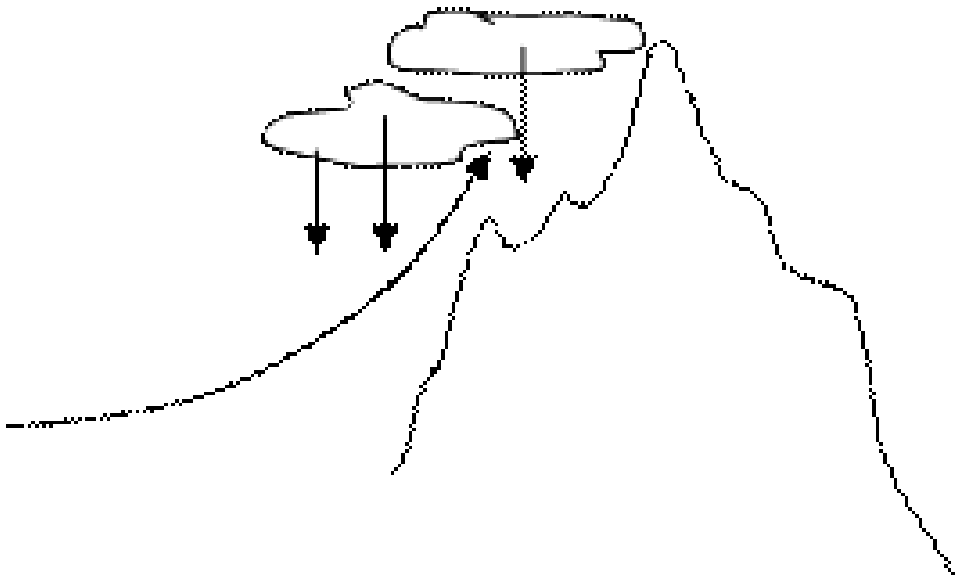
- Intense sunshine (typical of ocean)
- High humidity and temperature (typical of ocean)
- Low surface friction (typical of ocean)

ii. *Monsoons*: 180° reversal in prevailing wind direction with season. Partly due to differential heating of oceans and continents (India, West Africa), brings a season of wet weather. Becomes more prominent with topographic effects that creates a semi-permanent low pressure over land, thereby drawing moisture from the ocean.

AGRO-METEOROLOGY MEASUREMENTS

A. Creation of Saturation Conditions

4. Orographic Cooling (lifting): Cooling as air rises over mountain range. Greater percentage of precipitation occurring on windward side of mountains



AGRO-METEOROLOGY MEASUREMENTS

B. Change of Phase (nucleation):

Once saturated conditions exist droplets are formed, however surface tension resists nucleation. Therefore, the presence of small particles (aerosols e.g. sea salt, pollutants such as dust, combustion by-products and ice) provides the initial condensation nucleus for droplet formation.

C. Droplet Growth

Principal growth mechanism is coalescence or growth by collision. Collisions are generated by turbulent air motion and falling rain drops. A droplet must grow to a size large enough to overcome the upward convective velocities.

AGRO-METEOROLOGY MEASUREMENTS

Cloud Seeding

- Is the artificial nucleation to induce precipitation.
- Rainfall may be induced by the introduction of artificial nucleation materials (such as Silver iodide or dry ice) into the atmosphere.
- The long term effect of such activity on rainfall variability is NOT clearly known, as extreme events or events that would have occurred without seeding is difficult to evaluate

AGRO-METEOROLOGY MEASUREMENTS

Rainfall measurement

- Rain gauges are used to measure the depth and intensity of falling rain on flat surfaces.
- There are two main types of rain gauges namely:
  1. non-recording
  2. recording types.

AGRO-METEOROLOGY MEASUREMENTS

Rainfall data

The important parameters of rainfall are intensity, duration and frequency.

- The intensity is amount of rainfall per unit time expressed in mm per hour
- The time period for a given storm is referred to as the duration
- The period taken for a give storm of a certain intensity and duration to re-occur is referred to as the return period or recurrence interval.

Factors affecting rainfall measurement

- Design of the rain gauge
- Topography of location
- Presence of obstruction (such as trees and buildings)

AGRO-METEOROLOGY MEASUREMENTS

Non-recording rain gauge:

- It is used to record the rainfall amount over 24 h (day).
- The measurement is by a measuring cylinder or dip-stick.

Advantage:

- It is economical
- relatively free of maintenance.

Disadvantage:

- Require service after the rains

Main sources of error for non recording rain gauge:

- measuring stick
- leaks in the funnel or can
- denting of can
- evaporation.

AGRO-METEOROLOGY MEASUREMENTS

Rainfall measurement



Standard non-recording rain gauge

AGRO-METEOROLOGY MEASUREMENTS

Recording rain gauge

- It is use to record the rainfall amount with time on a continuous basis.
- It may be a weighing bucket, tipping bucket or float-siphon type.

AGRO-METEOROLOGY MEASUREMENTS

Types of recording rain gauge

1. Weighing Bucket rain gauge - weighs the amount of water collected and converts it to mm. The amount is recorded on paper, similar to the way it is done on the thermohygrograph. Newer versions record data in digital form on data loggers.
2. Tipping bucket - counts the number of tipping per unit time and with the volume of bucket known, the amount of rainfall can be calculated
3. Float-siphon – float rises with rainfall, giving an indication of rainfall amount

AGRO-METEOROLOGY MEASUREMENTS

Recording rain gauge

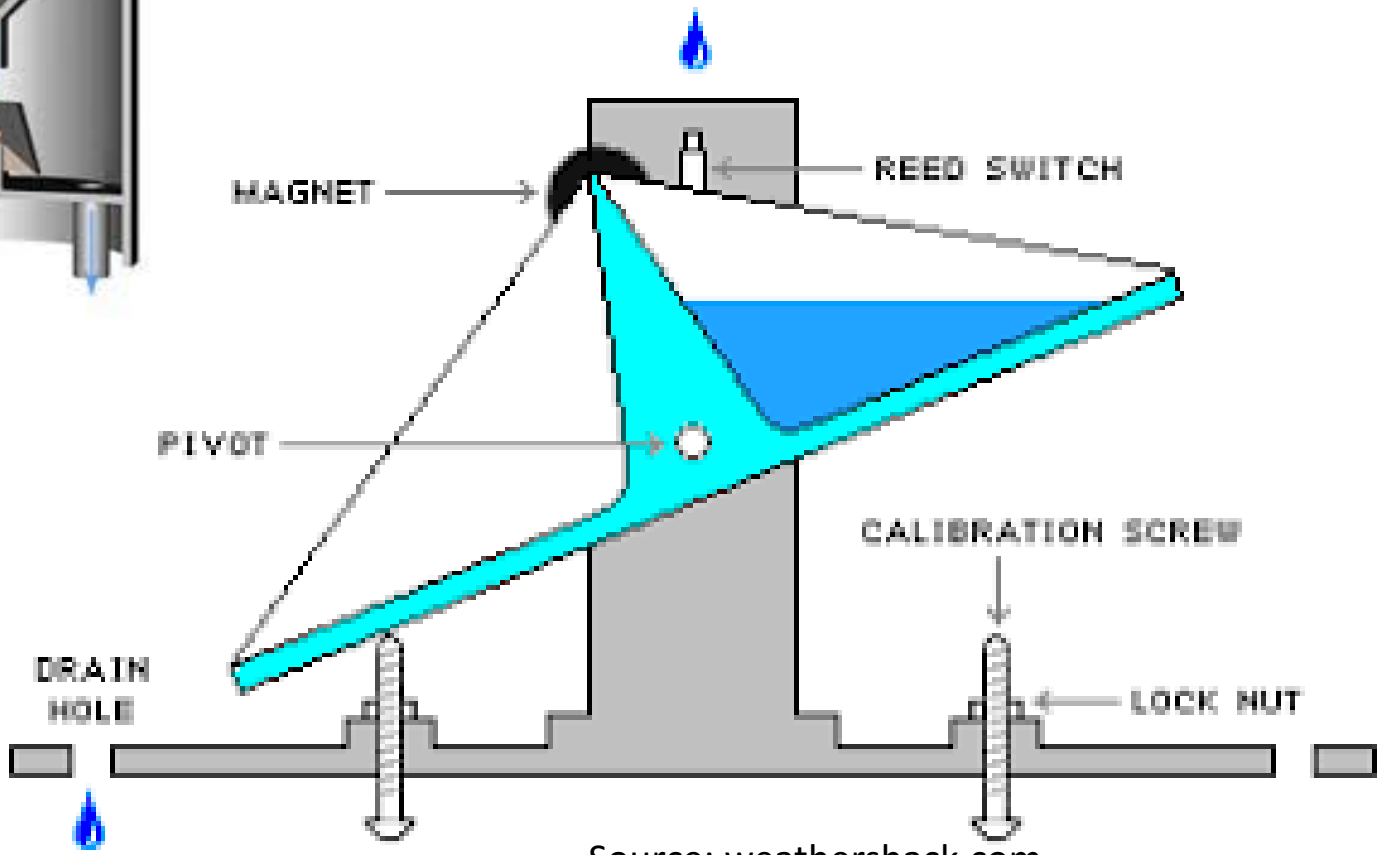




Tipping Bucket Raingauge



Source: bellflowsystems.co.uk



Source: weathershack.com

AGRO-METEOROLOGY MEASUREMENTS

Advantages of **recording rain gauge**:

1. Suitable for poorly accessible areas
2. Determination of rainfall intensity
3. Provide continuous rainfall data

AGRO-METEOROLOGY MEASUREMENTS

Effective rainfall

- This is the rainfall that infiltrates into the soil and is available for crop use.
- It is defined as that portion of total rainfall which is directly or indirectly useful for crop production.
- It must take care of the consumptive use by crops, rainfall for land soaking, land preparation, leaching of salts etc.
- The effective rainfall values used for irrigation system design and water scheduling varies in different tropical countries (i.e. 70-90 %).

AGRO-METEOROLOGY MEASUREMENTS

Effective rainfall

The actual effectiveness of rainfall depends on:

- Rainfall amount and intensity
- Evaporative demand
- Land and soil characteristics
- Antecedent soil-water status
- Groundwater status
- Farm and crop management practices
- Crop characteristics

TEMPERATURE

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- **Air temperature (T)**



Temperature

- Temperature is the degree of hotness or coldness of a body.
- For the atmosphere, the air temperature gives how warm the environment is.
- It is measured with the air thermometer which is placed in the Stevenson screen.
- In the screen there are 4 thermometers

Temperature

Thermometers in the Stevenson screen :

- Dry bulb thermometer: for direct reading of **air** temperature
- Wet bulb thermometer: for reading temperature under **moist** condition
- Maximum thermometer: is used for reading the **maximum** temperature over **24 hour period**
- Minimum thermometer: is used for reading the **minimum** temperature over **24 hour period**

Temperature

- **Dry-bulb Temperature:** The dry-bulb temperature of air is measured by a thermometer which is freely exposed to the air but is shielded from radiation and moisture.
- **Wet-bulb Temperature:** The Wet-bulb temperature of air is measured by a thermometer whose bulb is covered by a muslin sleeve which is kept moist with clean distilled water, freely exposed to the air and free from radiation.
- Using the dry and wet bulb the dew point, vapour pressure, and relative humidity can be estimated.

Temperature

Maximum thermometer:

- A maximum thermometer is a mercury-in-glass thermometer made with a constriction in the bore between the bulb and the graduated portion of the stem.
- Rising temperature forces a portion of the mercury into the graduated section.
- This mercury is retained and gives a reading of the highest temperature reached in 24h until reset.

Temperature

Minimum thermometer:

- A minimum thermometer is an alcohol-in-glass thermometer with a small freely moving, dark-colored glass index placed in the bore.
- As the temperature falls, the retreating upper end (meniscus) of the alcohol column moves the index toward the bulb.
- When the temperature rises, the index remains at the lowest point in 24h until reset.

Temperature

$$Mean\ daily\ temperature = \frac{T_{max} + T_{min}}{2}$$

- Temperature of a given location: is the average of the mean daily temperature over a period of 30 years

$$Daily\ temperature\ range = T_{max} - T_{min}$$

- Daily temperature range: is the difference between the highest and lowest temperatures recorded in a particular day.

Temperature

Seasonal temperature variations are influenced by:

- Angle of inclination of solar rays
- Distance between the sun and the earth
- Air mass movement (i.e. seasonal winds which contribute to rain)
- Period of the year
- Land cover type
- Land topography
- Prevailing weather situation (cloud cover, moisture, wind)

Temperature

Factors affecting temperature:

- Latitude
- Altitude (the height above mean sea level)
- Elevation (the height above a given land surface)
- Wind
- Contrast between land and water
- Warm and cold ocean currents
- Local impact such as slope and orientation

Temperature

Importance of temperature to organic life

1. Rate of physical and chemical processes within plants and subsequent control of biological reactions
2. The diffusion rate of gases and liquids varies with temperature
3. Solubility of different organic substances varies with temperature
4. Stability of enzyme and their reactions are affected by temperature

Soil Temperature

Soil thermometers

- Soil thermometers are specialized thermometers used to measure soil temperature when placed at the desired soil depth.
- In modern times some weather stations have automatic soil temperature measuring instruments.
- Thermistors and thermocouples are some of the standard soil temperature sensors that are available.

Soil Temperature

- Soil temperature varies in response to exchange processes that take place primarily through the soil surface.

- The effect of temperature are propagated into the soil profile by transport processes and are influenced by the specific heat capacity, thermal conductivity and thermal diffusivity.

Soil Temperature

Role of soil temperature:

- Influences soil chemical and biological interactions
- Seed germination: optimum soil temperature is required for good germination for a given crop seed
- Plant growth: optimum soil temperature range for good crop growth of any crop specie
- Crop yield: soil temperature must reach the optimum for a specific crop to get a good stand
- Soil temperature influences soil borne diseases like seedling blight, root rot etc.
- Decomposition of organic matter

Soil Temperature

Modification of soil thermal regime is by:

- use of tillage practices
- application of mulch
- shelter or shading.

- Mulches, depending on what they are and their albedo, will either warm or cool the soil.
  - A highly reflective material will lower the temperature by reducing the radiant energy reaching the soil.
  - A denser and less reflective material will increase the soil temperature by inhibiting evaporation.

SUNSHINE

Sunshine duration

Stokes sunshine recorder

- Consist of spherical glass which focuses the sun's rays on to a specially treated calibrated card where a trace is burnt or scorched.
- The accumulated burnt trace as the sun moves round during the day gives a measure of the total length of bright sunshine hours.
- The card has to be change daily at the observational time in the morning.

The amount of sunshine received at a given location depends on:

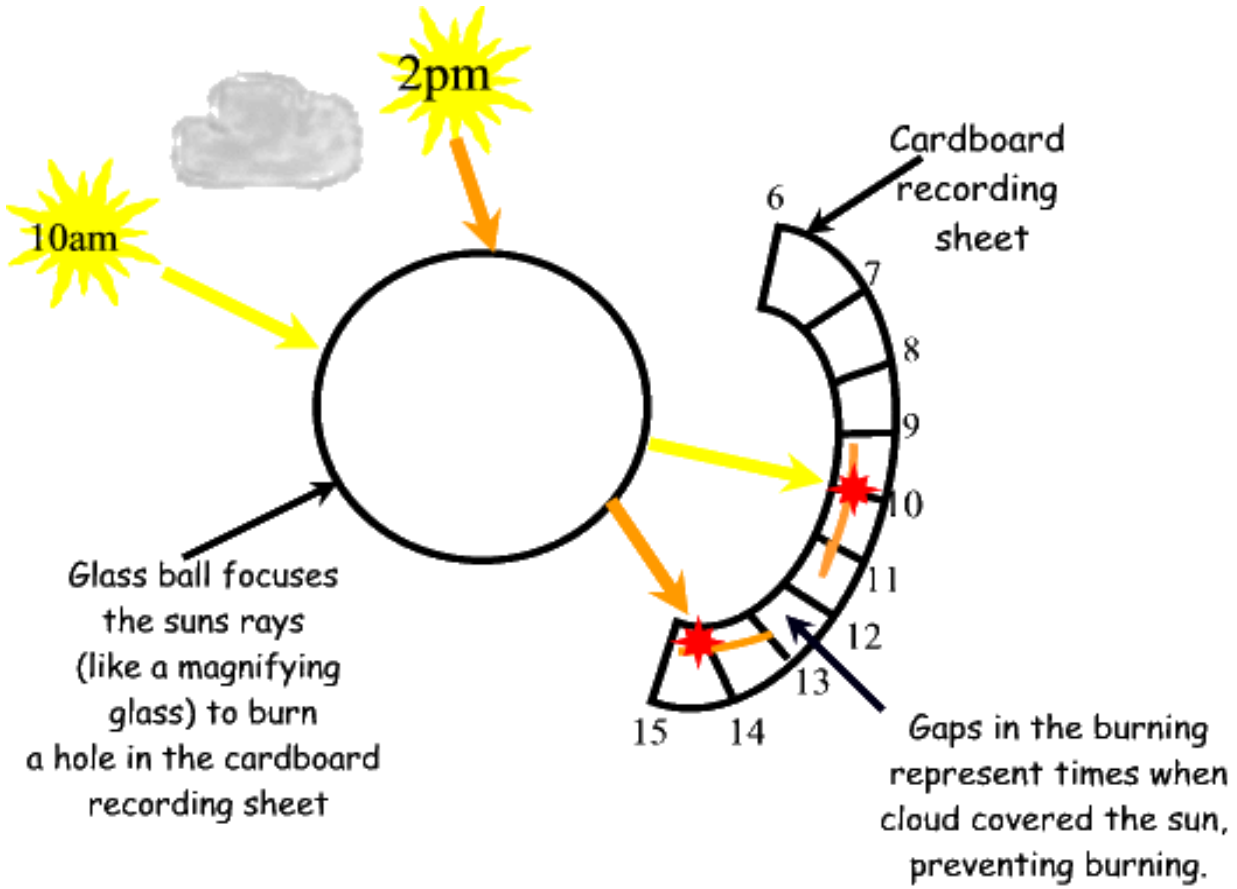
- Latitude
- Amount of cloud in the sky

Stokes sunshine recorder



Sunshine duration

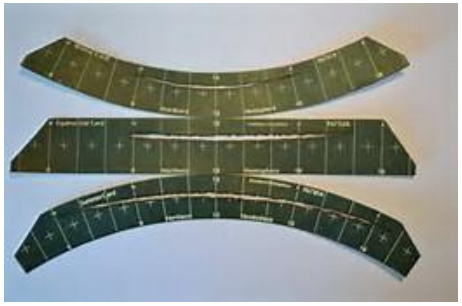
Stokes sunshine recorder

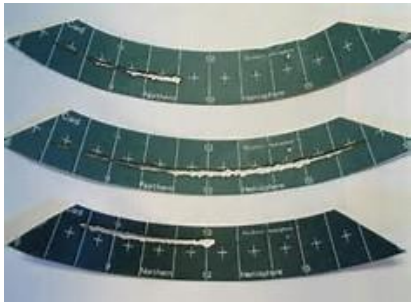


Sample sunshine recording sheets

- Sunrise
- Sunset
- Cloudy periods









HUMIDITY

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Humidity

- Humidity: is the amount of moisture in the air
- Common expressions for humidity or amount of moisture in the air in the field of meteorology:
  - vapour pressure
  - dew point depression
  - relative humidity

Vapour pressure

- The pressure exerted by water vapour if it exists alone at the temperature and pressure of atmospheric air is the vapour pressure.
- It is the partial pressure of water vapour.
- Vapour pressure increases with temperature
  - At higher temperature more molecules have the necessary kinetic energy to escape the attractive forces of the liquid phase
  - The more molecules in the vapour phase, the higher the vapour pressure

Characteristics of water vapour pressure

- At 100°C the vapour pressure of water is 760 mmHg (1 atm) or equal to the atmospheric pressure of the liquid (in an open container)
- At high altitudes (i.e. up in the Mountains) the air pressure is less than at sea level. Thus, water will boil at a lower temperature (i.e. vapour pressure needed to support a bubble is lower at high altitude).
- Water vapour acts as a green house gas by trapping infrared radiation reflected from the earth.
- Desert temperatures can become much lower at night, as there is little moisture in the air to trap the heat.

Saturated Air

- At any given temperature, a fixed volume of air has the capacity to absorb only a certain quantity of water vapour.
- When this capacity is reached, the air is said to be **saturated with water vapour**.
  - i.e. There is a limit to the amount of water vapour the air can hold at a given temperature.
- Air that is holding as much moisture as it can at a given temperature is called saturated air.
- Any moisture introduced into saturated air will condense.

Vapour Pressure Deficit/ Saturation Deficit

- Vapour pressure deficit is the difference between saturation vapour pressure and actual vapour pressure for a given period at the same temperature.
- It is the amount of additional water vapour that the air can hold.
- It is an indicator of the evaporative capacity of the air.

Dew Point Temperature

- The dew point temperature is the temperature at which the air can no longer hold all of its water vapour, and some of the water vapour must condense into liquid water.
- Dew point is the temperature at which water vapour saturates from an air mass into liquid or solid usually forming rain, snow, frost, or dew.
- Dew point normally occurs when a mass of air has a relative humidity of 100%. This happens in the atmosphere as a result of cooling through a number of different processes.
- Dew point temperature is defined as the temperature at which the air would have to cool (at constant pressure and constant water vapour content) in order to reach saturation.

Dew point depression

- It is the difference between the Temperature of the air and the dew point temperature.
- Low dew point depression values mean that the air is very moist and an increased likelihood of clouds forming.
- Larger dew point depression values mean drier air.

Humidity

Absolute Humidity (AH):

- amount of water vapour in a given volume of air (grams of vapour/cc of air)

Water Vapour Capacity (WVC):

- amount of water vapour that could be held in a given volume of air.
- It is dependent on temperature as warm air can hold more moisture than cold air.

Relative Humidity

- **Relative Humidity**: ratio of how much water the air holds to the amount that it could have, expressed in percentage

$$RH(\%) = \frac{Actual\ vapour\ presure at\ a\ given temp.}{Saturated Vapour\ pressure at\ the\ same temp.} \times 100$$

$$RH = \frac{AH}{WVC} \times 100$$

Characteristics of RH

- Relative humidity of 100% means the air can hold no more water (rain or dew is likely)
- A relative humidity of 0% indicates there is no moisture in the atmosphere.
- Warm air can hold more moisture than cold air. So, if warm air and cold air contain the same amount of moisture, the warm air will have a lower RH.
- Two conditions that can cause the relative humidity to rise: when the temperature falls or when moisture is added to the air.
- Relative humidity, combined with air temperature, can be used to estimate the actual amount of moisture in the atmosphere, sometimes referred to as precipitable water.

Relative humidity

Saturation:

- point at which evaporation is less than condensation
- RH = 100%
- AH = WVC

Problems or challenges of high relative humidity

- condensation and mold problems occur when the relative humidity is too high.
- The effect of moisture in gases also plays a very significant role in corrosion phenomena which can result in damage and loss of unprotected metals, like iron and steel structural components, and improperly treated or stored steel and other metal products.

Uses or Benefits of Relative humidity

- Meteorologists can use it to predict the weather
- Pathologists to predict disease development on plants, and
- Agricultural Scientists/Engineers to estimate evapotranspiration

Measurement of relative humidity

Direct measurement using

1. Hair hygrometers

- Uses human or horse hair as a sensing element.
- Changes in hair length in proportion to humidity of the air.
- The change in length is transmitted and magnified by a lever system into a measurement of relative humidity.
- Challenges:
  - response to changes in humidity is slow
  - not very dependable at very high RH



Measurement of relative humidity

Direct measurement using

2. Electric hygrometer

- Works based on substances whose electrical properties change as a function of their moisture content.
- As the humidity of the air around the sensor increases, its moisture content increases, proportionally affecting the sensor's electrical properties.
- Challenges:
  - Sensors lose their calibration if allowed to become contaminated or water condenses on them.

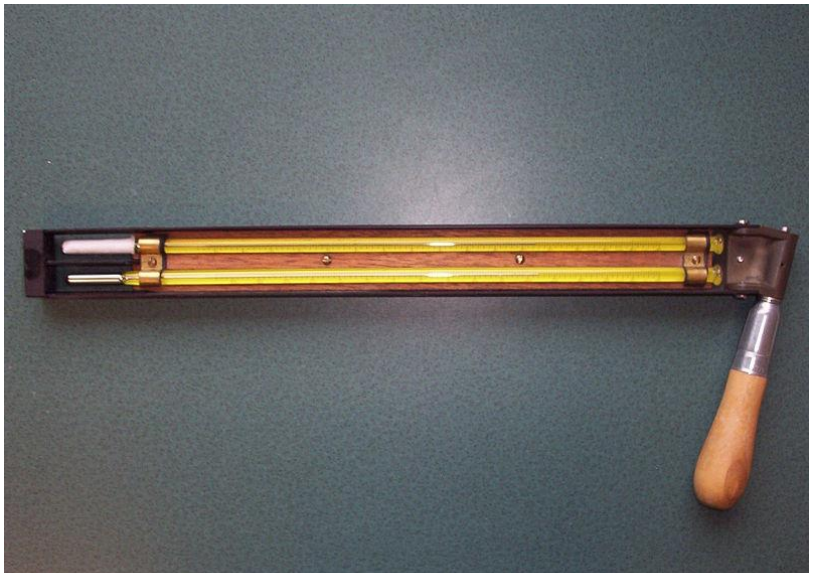
Measurement of relative humidity

Indirect measurement - Psychrometer

- Psychrometer is a type of hygrometer.
- It consists of two matched thermometers
- one of which measures air temperature T_d called the **dry-bulb**.
- The other thermometer is covered with a wet muslin cloth and reads the **wet-bulb** temperature T_w.
- Relative humidity can be determined from wet bulb and dry bulb temperatures.
- The difference in temperatures $T_d - T_w$ is proportional to relative humidity of the air.

Sling Psychrometer (1/3)

- The simplest hygrometer - consists of two thermometers mounted together with a handle attached on a chain.
- One thermometer is for dry bulb reading.
- The other has a cloth wick over its bulb i.e. the wet-bulb thermometer.



Sling Psychrometer (2/3)

- When a reading is to be taken, the wick is first dipped in water and then the instrument is whirled around.
- During the whirling, the water evaporates from the wick, cooling the wet-bulb thermometer.

Sling Psychrometer (3/3)

- Then the temperatures of both thermometers are read.
- If the surrounding air is dry, more moisture evaporates from the wick, cooling the wet-bulb thermometer more so there is a greater difference between the temperatures of the two thermometers.
- If the surrounding air is holding as much moisture as possible - if the relative humidity is 100% - there is no difference between the two temperatures.
- Meteorologists have worked out charts of these differences for each degree of temperature so that the observer can find relative humidity easily.

Humidity

- **Wet bulb depression** – the difference between the dry bulb and the wet bulb temperature readings is called the wet bulb depression.
- The wet bulb depression increases as the relative humidity decreases
- The lesser the wet bulb depression, the wetter the air
- The greater the wet bulb depression the drier the air

- **Thermohygrograph:** The hygrothermograph records both relative humidity and temperature on a graph

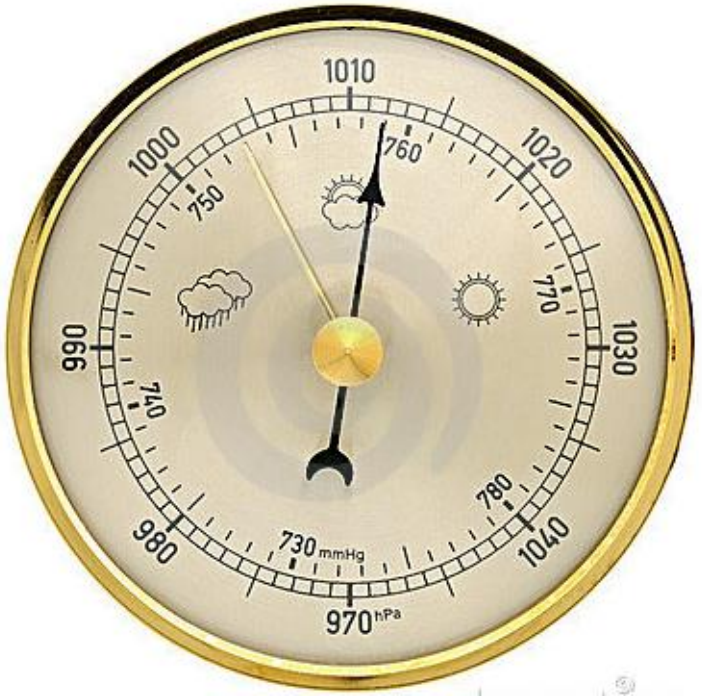
ATMOSPHERIC PRESSURE

Nana Kwabena

Atmospheric Pressure

- It is defined as the force per unit area exerted against a surface by the weight of the air above that surface.
- Atmospheric pressure is measured with an instrument called a barometer
- Atmospheric pressure is also referred to as barometric pressure –

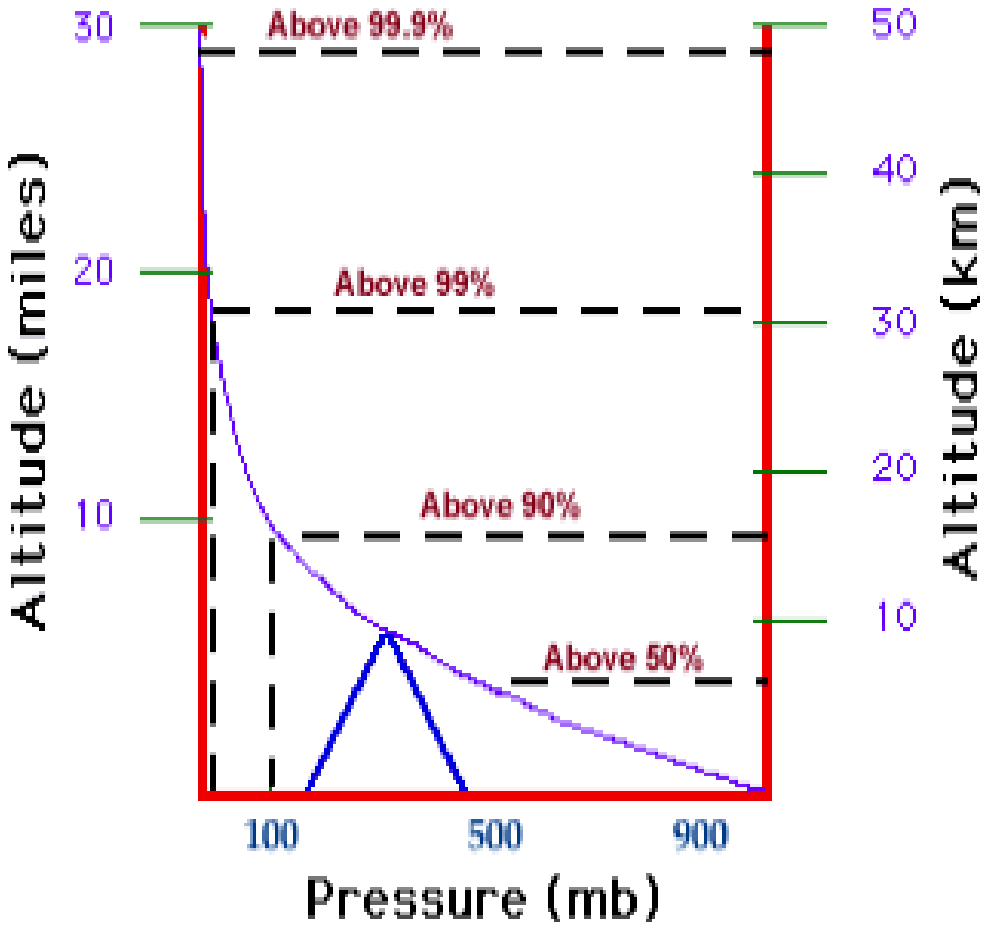
  millibars (mb), hPa or mmHg



Atmospheric Pressure

- Since the number of air molecules above the earth surface decreases with height, pressure likewise decreases with height.
- For example, there are fewer air molecules above the 50 kilometre (km) surface than are found above the 12 km surface.
- Atm. pressure decreases with increasing altitude.

Atmospheric Pressure



Changes in Atm. Pressure with Altitude

Atmospheric Pressure

- **Isobars** are lines of constant pressure.
- A line drawn on a weather map connecting points of equal pressure is referred to as an "**isobar**" and is given in millbars

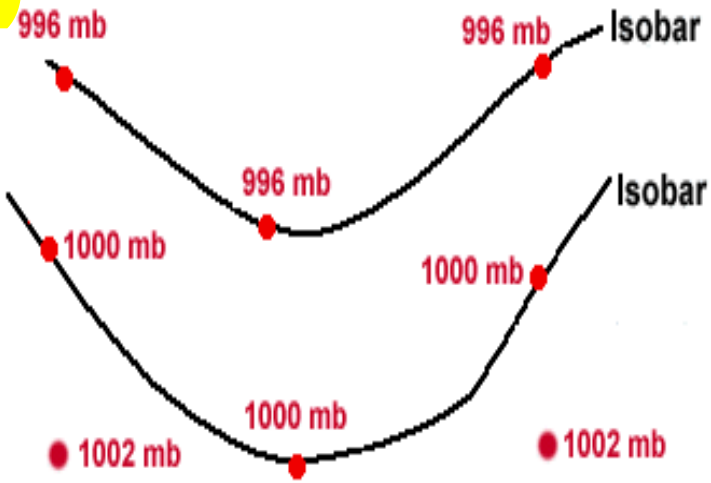


Fig.: Illustration of isobars

WIND

Wind speed and direction

- Wind results from a horizontal difference in air pressure
- This is due to the sun heating different parts of the Earth differently, causing pressure differences, the Sun is the driving force for most winds
- Atmospheric pressure closely determines the weather

Wind speed and direction

- Regions with pressure that are slightly higher or slightly lower than the mean atmospheric pressure develop as air circulates around Earth
- The air moves from regions of high pressure to low pressure, causing winds.
- The properties of the moving air (cool or warm, dry or humid) will determine the weather for the areas through which it passes
- Knowing the location of high and low pressure areas is vital to weather forecasting.

Wind Speed Measurement: Anemometers

- Wind is a two dimensional vector quantity expressed by its speed and direction
- The **cup anemometer** fixed on a 2m pole from the ground is used to record the instantaneous wind speed (ms^{-1}) or the wind run (i.e. distance in km)
- The cup anemometer is mounted on a vertical axis with three cups which capture the wind
- The number of revolutions per minute is registered electronically
- Wind speed is slowest at the surface, and increases with height until a constant mainstream value is reached at a certain height above the surface
- Wind is stronger over the ocean than land because the land surface is rougher than water surfaces.

Cup Anemometer





Wind Speed Measurement: Anemometers

- Anemometers are placed at a chosen standard height in an open field location.
- Anemometers are located in the field at a point at least 10 times the height of the nearest obstruction.
- In meteorology, the standard height is 10 m. In agro-meteorology the standard height is usually 2 m.
- If the wind speed is measured at a height other than 2 m, then the observed values are converted to 2 m height wind speed, using the following relationship:

$$U_{2m} = \frac{4.868}{\ln\left(67.75\,z - 5.42\right)} U_z$$

Where z – the height of wind speed, U_z measurement at z (m) above local ground level

Effect of Wind Speed

- Wind erosion
- Spread of forest fires, disease and insect pest infestation
- Damage to crops and buildings (property)
- Transport of water vapour from an evaporating surface
- Drive ocean currents
- Generate surface waves in oceans

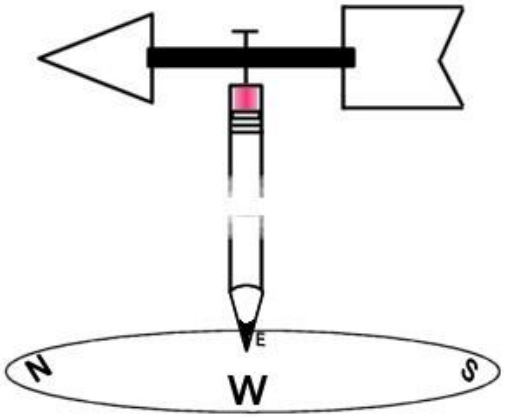
Wind Direction Measurement

- **Wind direction** is the direction from which a wind originates or comes from.
- It is measured using a Windsock or Wind vane
- Normally, anemometers are fitted with a **wind vane** to detect the wind direction.
- Windsock/vane points in the direction opposite from which the wind is blowing.



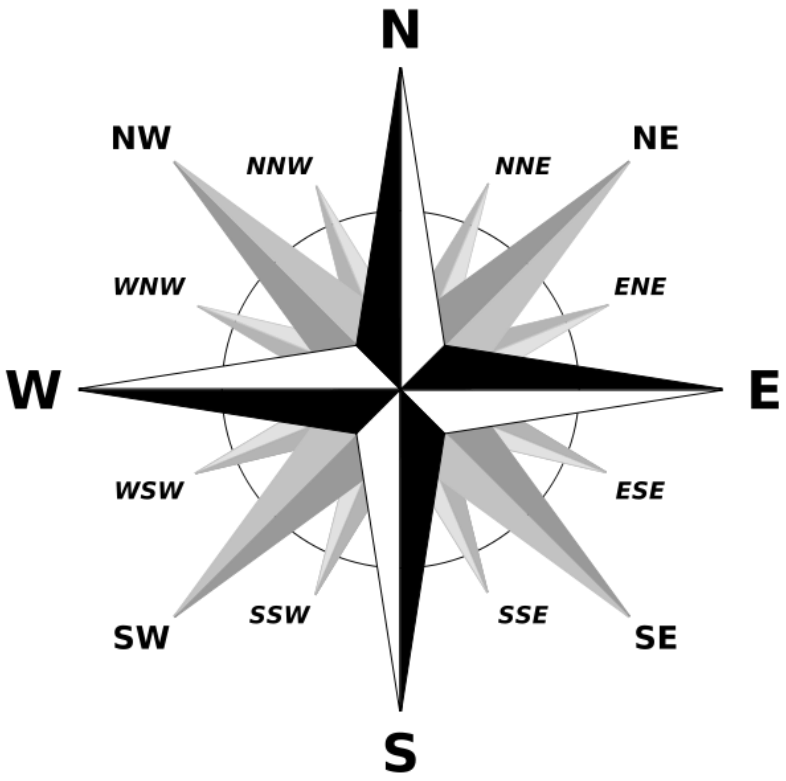
Windsock



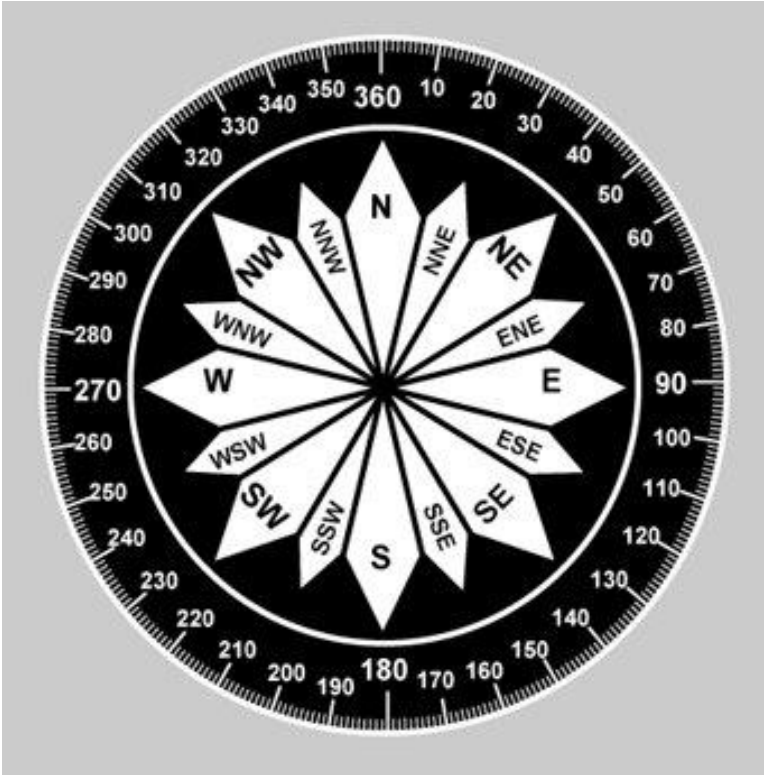




Wind vane



A Compass rose showing the four cardinal directions



A 16-point compass rose/windrose, showing correspondence with 360° circle

Visibility

- Meteorological visibility by day is the greatest distance at which a black object of suitable dimension located near the ground can be seen and recognized
- Meteorological visibility refers to transparency of air
- Visibility affects all forms of traffic: road, sailing and aviation.

Radiation

Radiation

The solar radiation at the earth surface is composed of three components.

These are:

- Direct
- Diffuse
- Reflected

Radiation measuring instruments

- The generic term for all radiation measuring instruments is radiometer.
- Radiation measuring instruments are classified according to their use.

1. **Pyrheliometer** measures the direct solar beam. Its sensing element must be pointed at the sun.
2. **Pyranometer** is used to measure global solar radiation, so it must respond to both the direct solar radiation and diffuse sky radiation. The sensing element must be a horizontal flat surface
3. **Pyrgeometer** is used to measure global longwave earth radiation

Radiation measuring instruments

4. **Pyrradiometer** measures total global radiation, both shortwave and longwave.

   Net pyrradiometer measures the difference between incoming and outgoing radiation

5. **Albedometer** measures the difference between incoming and reflected shortwave radiation (solar radiation)

Evapotranspiration

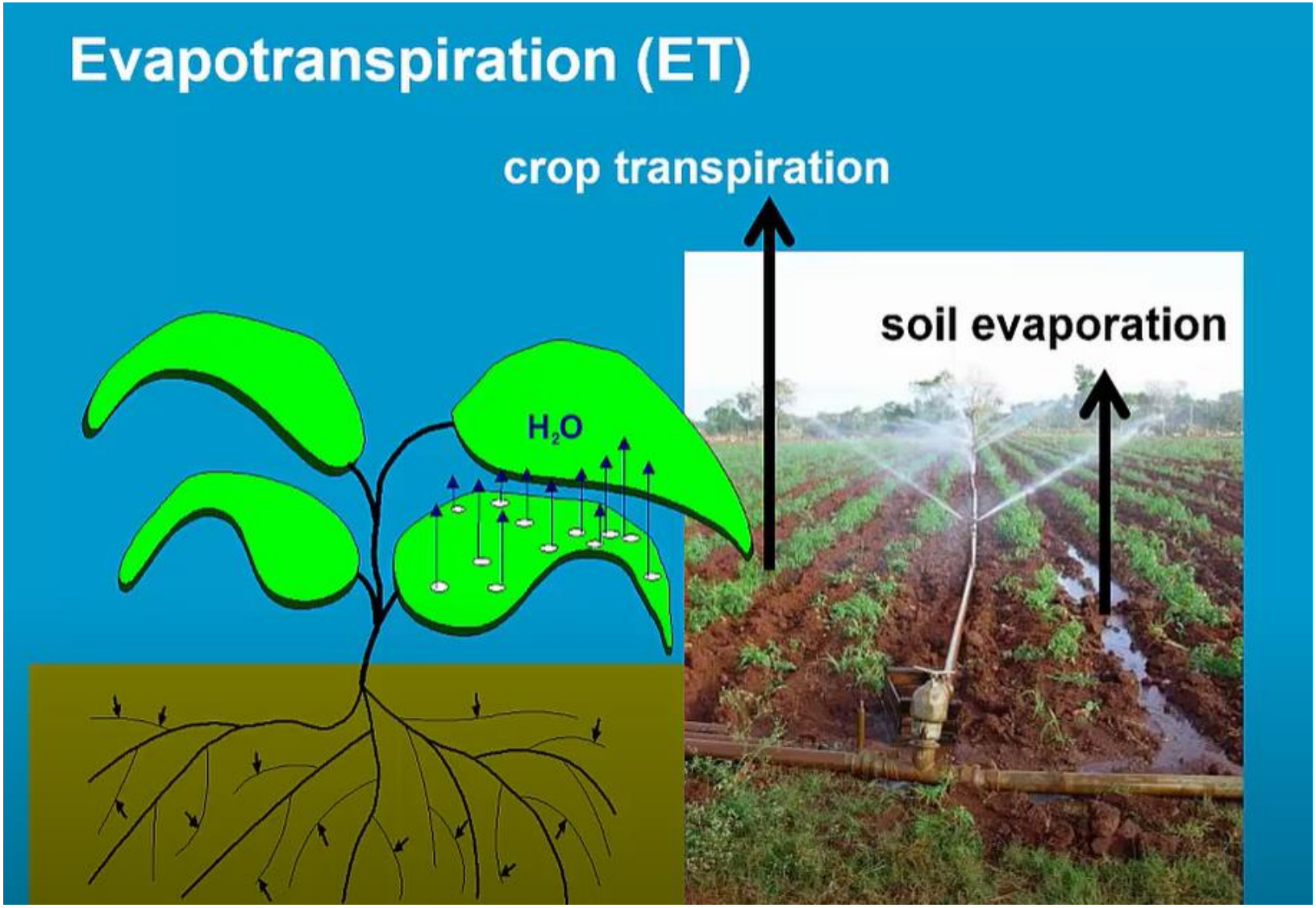
Nana Kwabena

Evapotranspiration

Evaporation: Is the transfer of water vapour into the atmosphere from an evaporation surface.

Evaporation surfaces: plant, soil and water surfaces.

Transpiration: is the transfer of water which has been transmitted from the soil through the plant from the plant surfaces.


Evapotranspiration (ET)
crop transpiration
soil evaporation
H_2O

Evapotranspiration

Evapotranspiration: Is the transfer of water from moist surfaces and living plant into the atmosphere (i.e. evaporation + transpiration)

Evaporation power: Is the ability of air to transform a unit mass of water into vapour and diffuse it.

Evapotranspiration

Conditions/requirement for evapotranspiration to occur:

1. Supply of heat to provide the latent heat of vaporization

2. Vapour pressure in the overlying air must be maintained at less than that at the evaporation surface (i.e. vapour pressure gradient or less moisture in the atmosphere compared to the evaporating surface)

3. Water/moisture must continuously be available for evaporation

Estimating evapotranspiration

Three (3) groups of methods-

1. **Mass transfer**:

It is assumed that water moves away from evaporating and transpiring surfaces in response to the combined phenomenon of turbulent mixing of the air and vapour pressure gradient.

This approach requires the measurement of wind velocity and humidity at two elevations and it is rarely practical.

Estimating evapotranspiration

2. **Energy balance**: Heat is required for evaporation, therefore if there is no change in water temperature then the net radiation or heat supplied is a measure of evaporation. This approach is more practical. Eg. Penman equation
3. **Empirical methods**: This is based on the assumption that the energy available for evaporation is proportional to the temperature. Eg. Blaney-Craddle

Pan Evaporation

- An **evaporation pan** is used to hold water for the determination of the quantity of evaporation at a given location.
- Such pans are of varying sizes and shapes, the most commonly used being circular or square
- Most common of the pans is the “Class A evaporation pan”
- Others are the "Sunken Colorado Pan“, in Europe, India and South Africa, a Symon's Pan (or sometimes Symon's Tank)
- Evaporation pans can be automated with water level sensors

Pan Evaporation

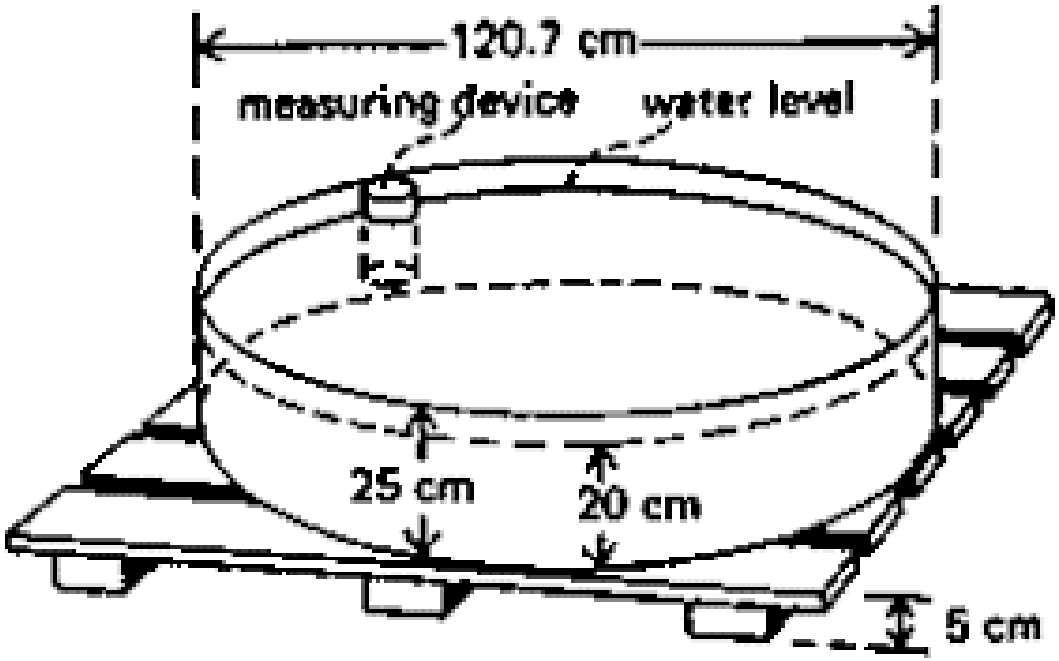
Class A pan

Characteristics

- Material type - galvanized pan
- Material thickness - 0.8 mm
- Pan diameter - 1.21 m
- Pan depth - 25.5 cm
- Pan colour - grey

- **Placement** - mounted on a wooden stand (5-15 cm above ground level) such that air freely circulates around and under the pan placed in an open space
- **Measurement** - A ruler (micrometer in a stilling well) is used to measure the drop in water.

Class A Pan



Class A Pan



Class A Pan



Pan Evaporation Calculation

Day/Date	Reading from pan (regular interval eg. one reading every 3 days	Rain (mm)	Evaporation power of the average reading per 3 day-period
Tue.			
Wed.			
Thurs.	27		27/3=9.0mm/day
Fri.			
Sat.		9	
Sun.	25		(25+9)/3= 11.3mm/day
Mon			
Tue.			
Wed.	29		29/3=9.7mm/day

Total evaporation for the 9 days: 81mm + rain

Average daily Evaporation: (81+9)/9 =10.0 mm/day

Pan Evaporation Calculation

Evaporation on rainy days: The rainfall depth is added to the reading on the pan gauge.

Example: If the daily pan reading indicates 6 mm and the rainfall recorded for the same day is 8 mm then the evaporation for that day is 14 mm

Limitation: On heavy rainy days the Class A Evaporation pan will likely overflow.

Factors that influences evapotranspiration (1/2)

1. Degree of soil water saturation
2. Soil texture and structure, which influences capillary rise and retention
3. Net radiation
4. Soil temperature
5. Air temperature surrounding the soil surface and the leaves
6. Saturation deficit

Factors that influences evapotranspiration (2/2)

7. Wind speed
8. Roughness of evaporating surface
9. Root and leaf development, as it determines the capacity of the plant to take-up water for transpiration
10. Salt concentration of the water as evapotranspiration decreases with increase in salt concentration
11. Elevation: Evaporation increases with elevation

Terminologies (1/2)

- **Potential evapotranspiration**:
  - Is the maximum possible amount of evaporation and transpiration that takes place given unlimited supply of moisture under a given meteorological condition.
  - That is the amount of water returned to the atmosphere by cultivated and uncultivated plants in optimal growing and watering condition.
- **Actual evapotranspiration**:
  - Is the evapotranspiration when water supply is not adequate.

Terminologies (2/2)

Reference evapotranspiration (ETo): is the potential evapotranspiration (ET) from a grass (e.g. alfalfa) reference surface. It is the maximum evapotranspiration of:

a. an extensive green grass cover
b. uniform height (8 -15 cm)
c. completely shading the ground
d. actively growing
e. no shortage of water

Methods of determining reference evapotranspiration (ETo)

- Blaney and Criddle
- Penman ETo according to Frere and Popov.
- Doorenbos and Pruitt
- Penman-Monteith

Compiled by NKay Study

AE 153 – Agricultural Meteorology & Climatology (2,1,2)

E. Bessah

Nana Kwabena

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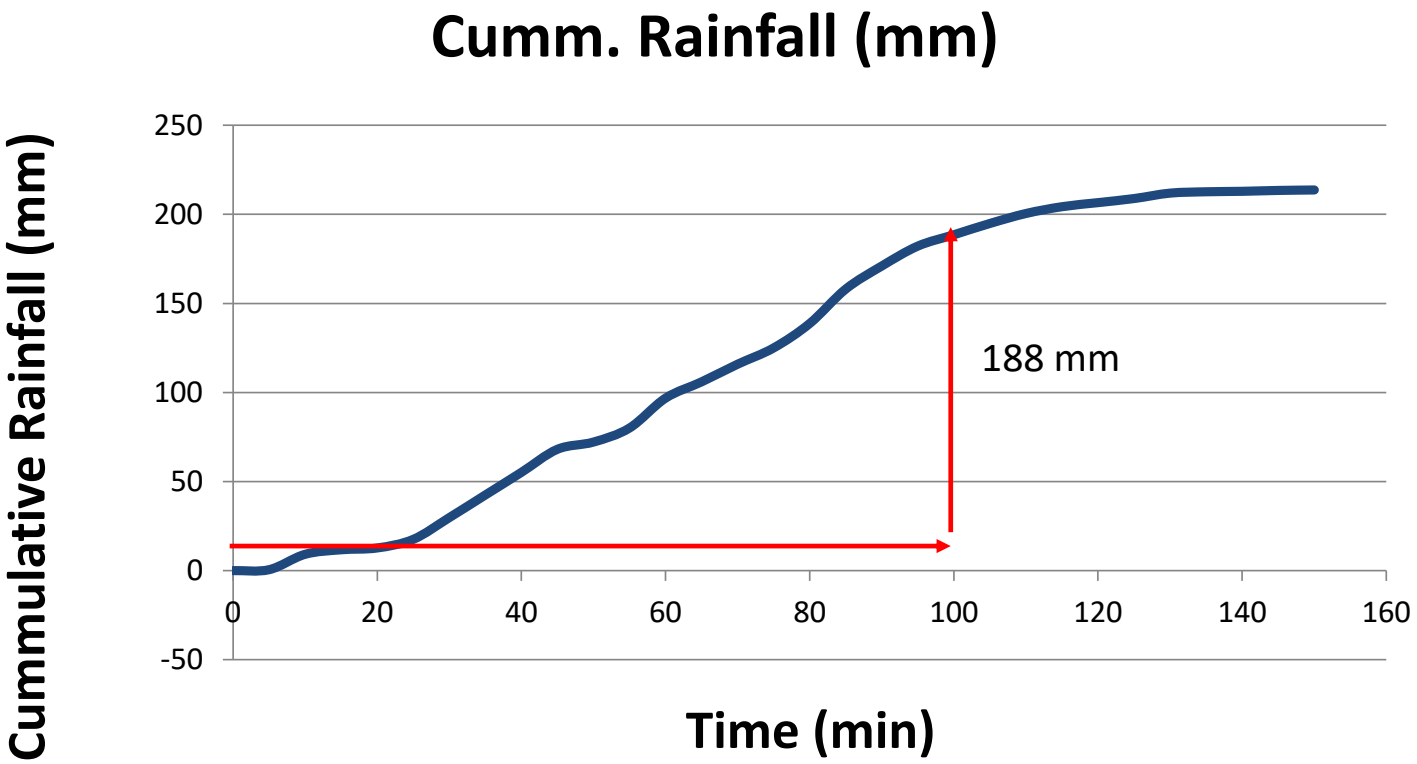
Data Presentation and Processing

Nana Kwabena

Rainfall data presentation

1. Cumulative mass curve

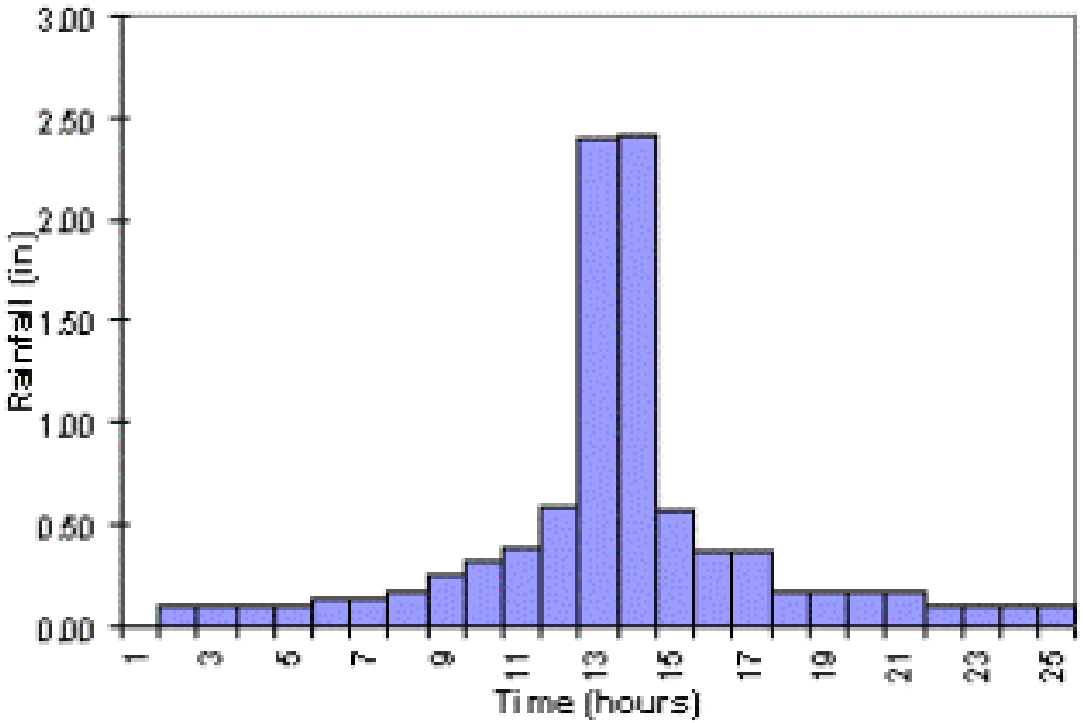
Plot of summation of rainfall increments as a function of time
Typically the output from a recording gauge is in the form of cumulative rainfall over time.



Rainfall data presentation

2. Rainfall hyetograph

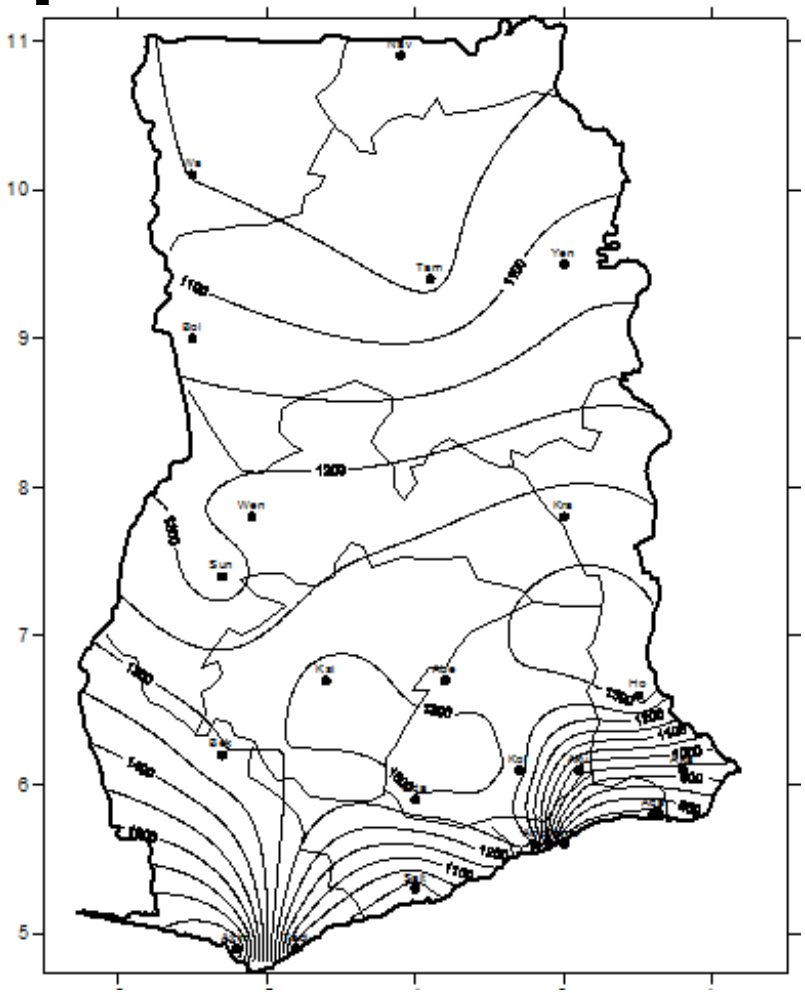
The depth or intensity of rainfall plotted against time interval to generate a hyetograph characterizing the time distribution of a storm or selected time period. Important input to hydrologic models (surface hydrology models).



Rainfall data presentation

3. **Isohyetal maps**

   Isohyetal map is a plot of constant rainfall depth/amount (sometimes intensity) contours based on rainfall data recorded at gauged locations (points). These are generated similarly to topographic contour maps. Isohyetal maps are used to represent long-term averages or individual storm events.



Isoyetal map of Ghana

Point rainfall data assessment

We measure rainfall at a point which is used for the following:

- Estimate average rainfall over an area (watershed)
- Transform point rainfall to aerial rainfall for model applications to large areas
- Estimate variation in rainfall data in terms of space or time
- Estimate missing record of gauge data or estimate data at an un-gauged location

Estimation of Precipitation over an Area

Methods: Arithmetic average/mean; Thiessen polygon; and Isoyetal

1. Arithmetic Averages*: Used when*

- *gauges are uniformly distributed*
- *individual gauge measurements do not vary greatly (typically for small areas)*

$$Pavg = \frac{\sum_{i=1}^{n} P_i}{n}$$

P_i = rainfall at gauge or location **i**

n = number of gauges or locations

2. Thiessen Polygon

- Underlying concept is that rainfall for an area is represented by the nearest gauge
- This is done by constructing polygons around each gauge, as follows:
  a. First, by drawing lines to connect each gauge.
  b. Then draw perpendicular bisector on each connecting line.
  c. Connect bisectors to form polygons
  d. Determine area for each polygon

Thiessen Polygon

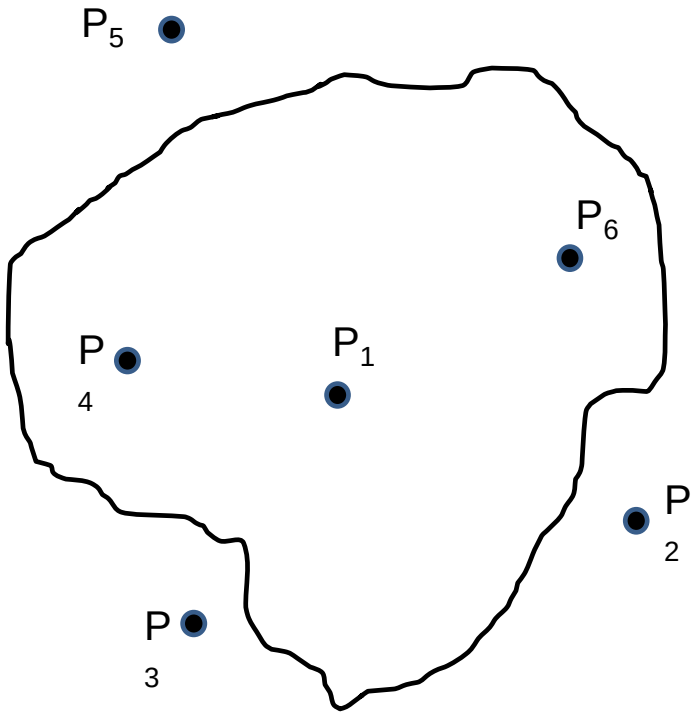
$$Pavg = \frac{\sum_{i=1}^{n} A_i P_i}{A_T}$$

n = number of polygons (gauges or locations)
A_T = total area
A_i = area of polygon around gauge or location i
P_i = rainfall at gauge or location i

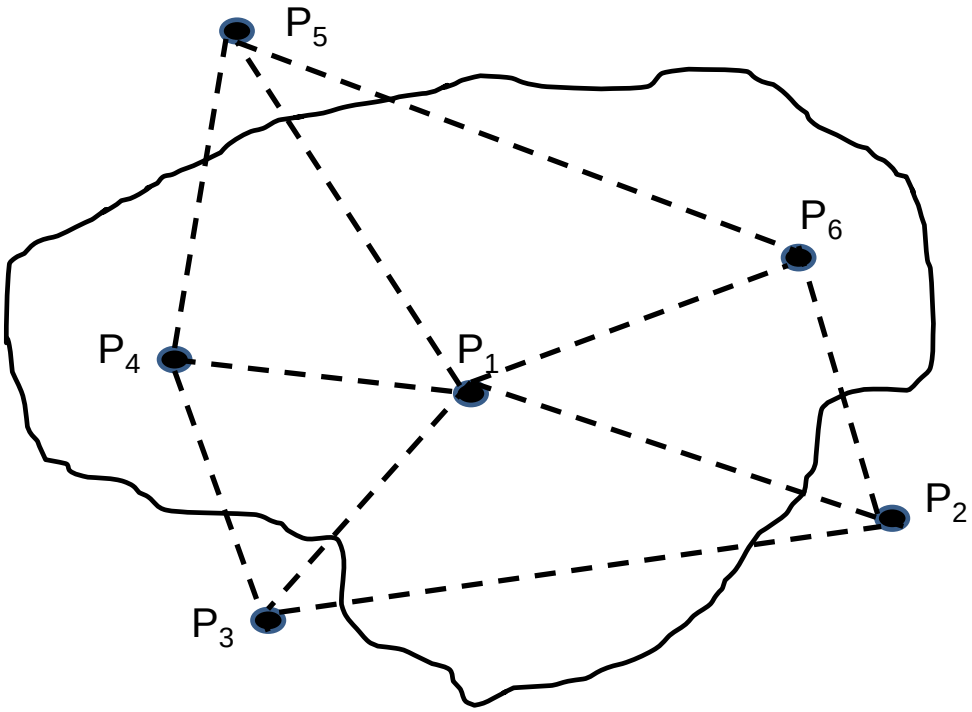
Sample calculation: Arithmetic Average and Thiessen polygon

Determine the mean rainfall using the arithmetic average and the Thiessen polygon methods

Rain gauge	**Precipitation (P_i)(mm)**
P_1	56
P_2	40
P_3	44
P_4	52
P_5	48
P_6	52
Total	

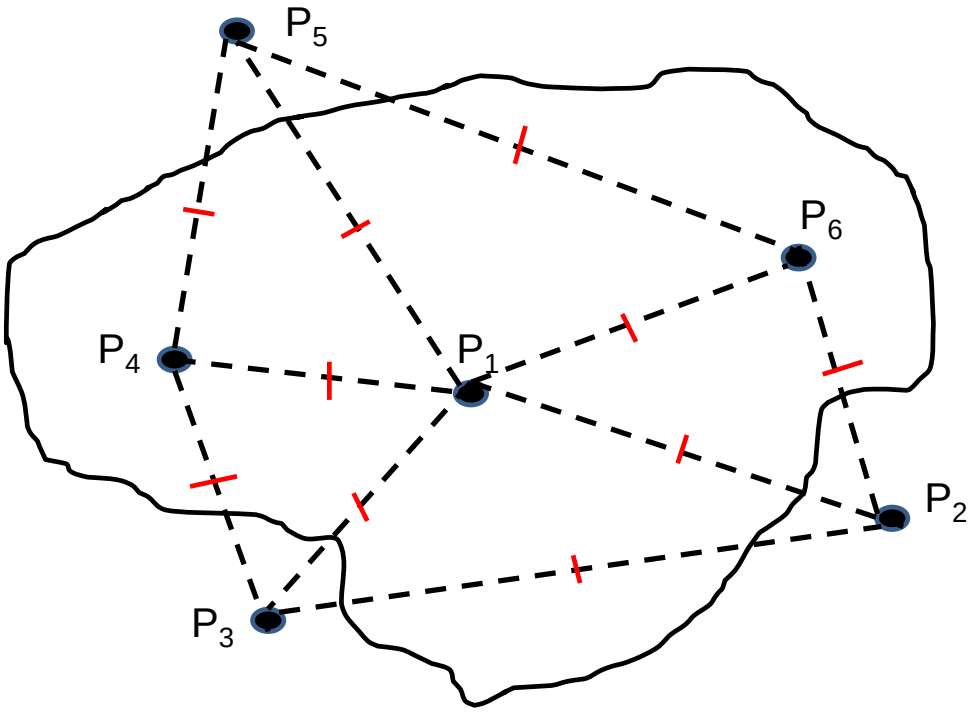


Thiessen polygon



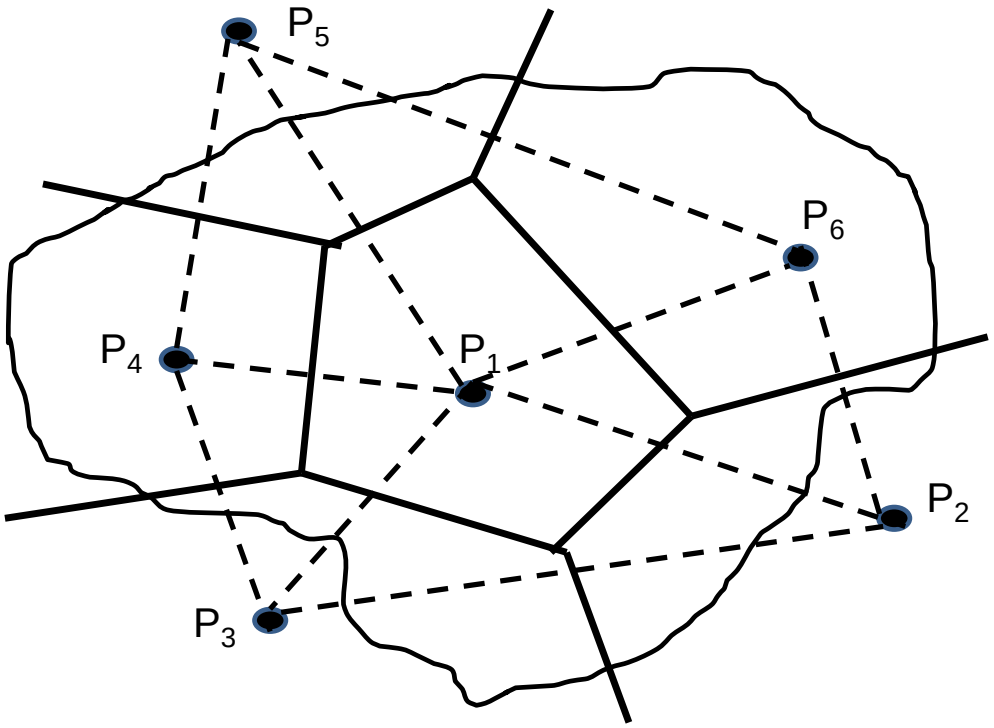
a. Drawing lines to connect each gauge

Thiessen polygon



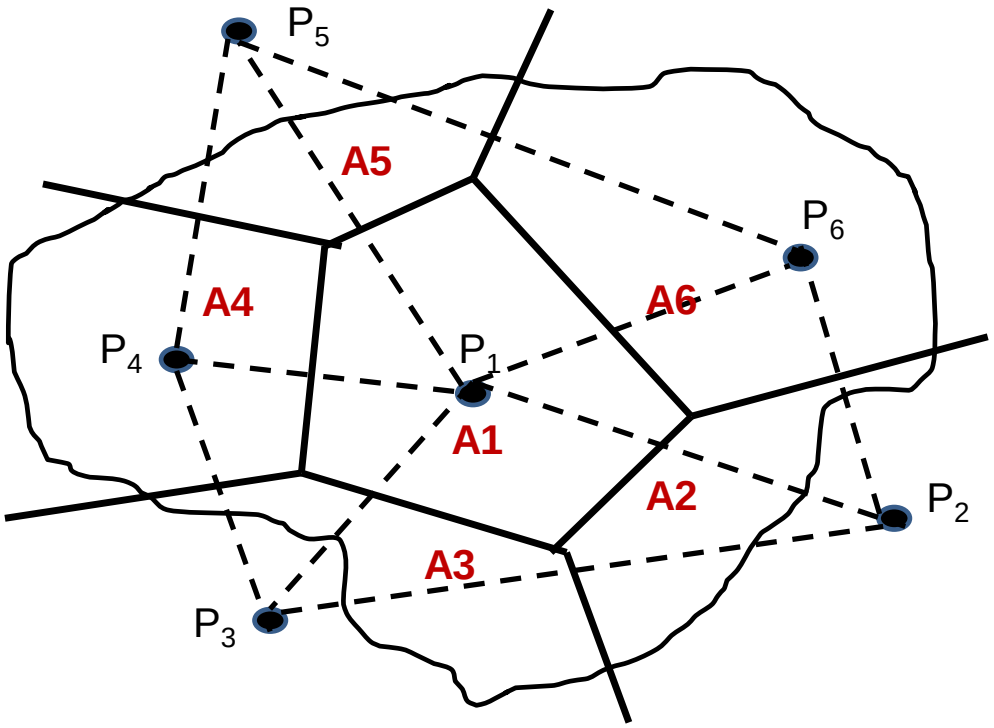
b. Then draw perpendicular bisector on each connecting line

Thiessen polygon



c. Connect bisectors to form polygons

Thiessen polygon



d. Determine area for each polygon

Arithmetic Average and Thiessen polygon

Rain gauge	**Precipitation (P_i)(mm)**	**Area of influence (A_i)(ha)**	**P_iA_i**
P_1	56	260	14560
P_2	40	58	2320
P_3	44	50	2200
P_4	52	250	13000
P_5	48	62	2976
P_6	52	290	15080
Total		970	50136

$$Pavg = \frac{\sum_{i=1}^{n} P_i}{n}$$

$$Av = \frac{292}{6} = 48.67$$

$$Pavg = \frac{\sum_{i=1}^{6} A_i P_i}{A_T}$$

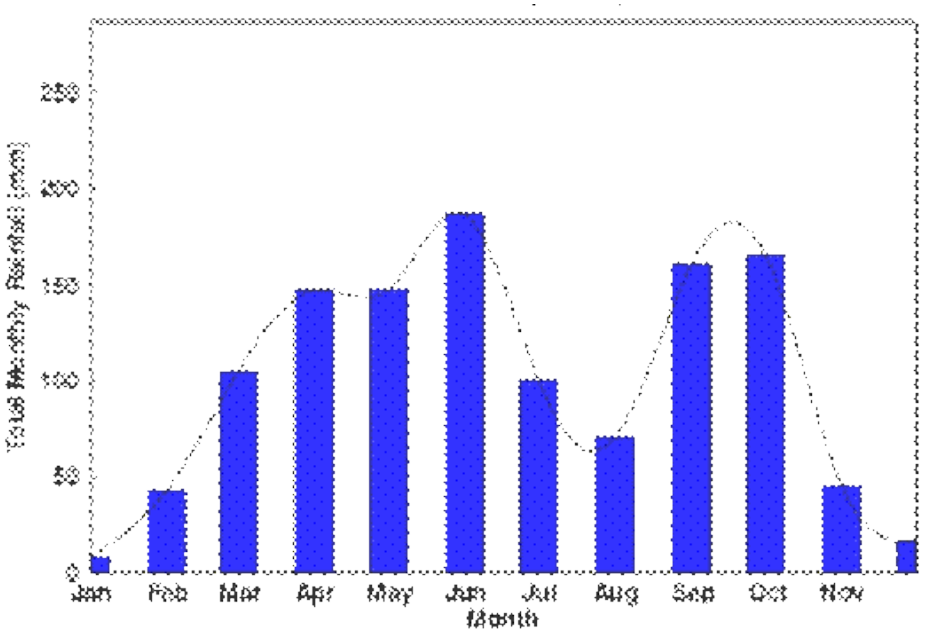
$$Av = \frac{50136}{970} = 51.69\ mm$$

3. Isohyetal Method

- Construct isohyets between gauges
- Calculate area between isohyets
- Estimate average rainfall between isohyetals and apply the same summation formula as in Thiessen Polygon where A_i and P_i represent the area and average rain between two isohyets.

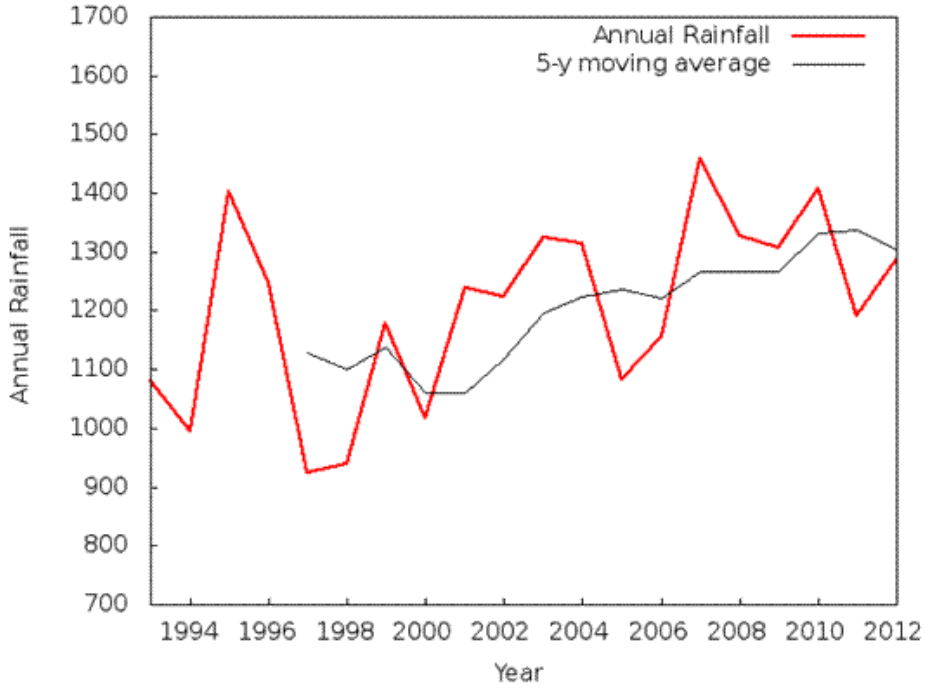
Rainfall Variability

Monthly rainfall distribution



Rainfall Variability

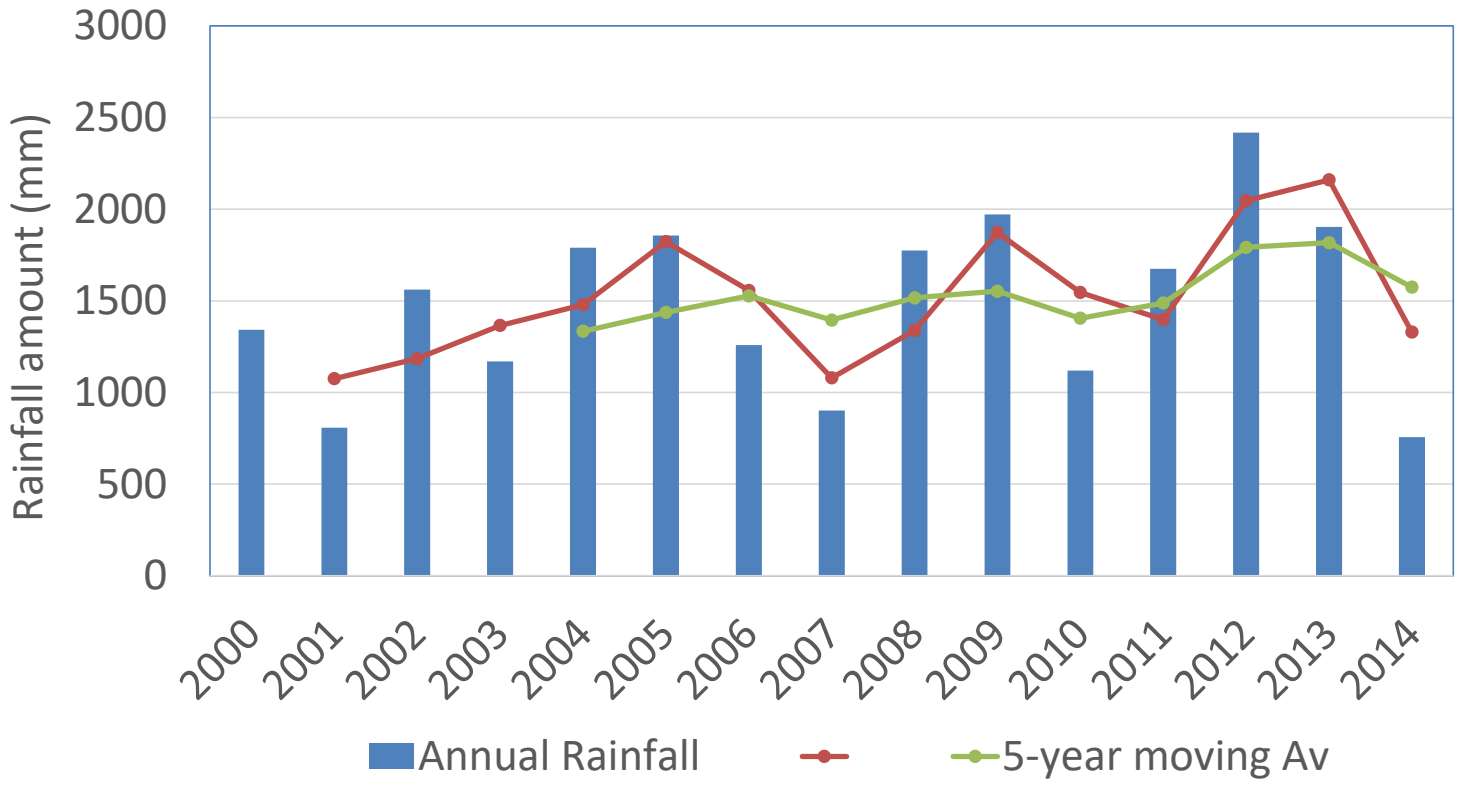
Rainfall trend analysis - Moving average i.e. average over number of months/years/events



Moving Average

SN	Year	Annual Rainfall	2-year moving Av	5-year moving Av	
1	2000	1342			
2	2001	808	1075		
3	2002	1562	1185		
4	2003	1169	1365.5		
5	2004	1790	1479.5	1334.2	
6	2005	1856	1823	1437	
7	2006	1258	1557	1527	
8	2007	902	1080	1395	
9	2008	1775	1338.5	1516.2	
10	2009	1971	1873	1552.4	
11	2010	1120	1545.5	1405.2	
12	2011	1675	1397.5	1488.6	
13	2012	2418	2046.5	1791.8	
14	2013	1903	2160.5	1817.4	
15	2014	756	1329.5	1574.4	
		1487			

Moving Average



Rainfall Variability

- **Mean** ($\bar{x}$) of data – average of the data

$$\bar{x} = \frac{\sum X_i}{n}$$

- **Standard deviation** (s) - measures the spread or variation of the individual rainfall data.

$$S = \sqrt{\frac{\sum (X_i - \dot{x})^2}{n-1}}$$

Rainfall Variability

Standard deviation

- The value of S is used to compare rainfall amounts variability for a given period for different stations.
- The smaller the value of S, the smaller the variation of rainfall.

Standard deviation estimation

A rough estimate of S is obtained by dividing the range (Highest minus lowest rainfall value) by a value depending on the number of data points

$$S = \frac{Xmax - Xmin}{n^*}$$

Number of data (n) closest to this given number	Divide range by (n*)
5	2
10	3
25	4
100	5

Rainfall Variability

- Coefficient of variation: measure of variation in data expressed as a percentage of the mean value. Useful in comparing rainfall variability of different stations

$$CV = \frac{s}{\bar{x}} \times 100$$

- CV < 0.20 or 20% → moderate variability and for most purposes an acceptable length of data series
- CV > 0.25 or 25% → either the series of data are too short for reliable estimates or the rainfall is highly variable

Sample calculation

Provided is May rainfall data for a 20 year period. Estimate the mean, standard deviation and CV, and comment on your answer

SN	Year	x
1	1981	85
2	1982	90
3	1983	65
4	1984	50
5	1985	40
6	1986	28
7	1987	65
8	1988	40
9	1989	82
10	1990	35
11	1991	30
12	1992	70
13	1993	75
14	1994	45
15	1995	55
16	1996	80
17	1997	68
18	1998	44
19	1999	50
20	2000	72

Sample calculation

SN	Year	X	$(X - \dot{X})$	$(X - \dot{X})^2$
1	1981	85	26.55	704.90
2	1982	90	31.55	995.40
3	1983	65	6.55	42.90
4	1984	50	-8.45	71.40
5	1985	40	-18.45	340.40
6	1986	28	-30.45	927.20
7	1987	65	6.55	42.90
8	1988	40	-18.45	340.40
9	1989	82	23.55	554.60
10	1990	35	-23.45	549.90
11	1991	30	-28.45	809.40
12	1992	70	11.55	133.40
13	1993	75	16.55	273.90
14	1994	45	-13.45	180.90
15	1995	55	-3.45	11.90
16	1996	80	21.55	464.40
17	1997	68	9.55	91.20
18	1998	44	-14.45	208.80
19	1999	50	-8.45	71.40
20	2000	72	13.55	183.60

Sample calculation

- $\dot{X} = \frac{\sum x}{n} = \frac{85+90+65+\cdots+72}{20}$
- $\dot{X} = \frac{1169}{20} = 58.45$
- $S = \sqrt{\frac{\sum(X_i - \dot{x})^2}{n-1}} = \sqrt{\frac{6998.95}{19}} = 19.19$
- rough estimate of s = 15.15

- $CV = \frac{s}{\bar{x}} \times 100 = \frac{19.19}{58.45} \times 100 = 32.83\%$

Sample calculation

- Rainfall at the station is highly variable therefore the data may be unreliable and /or the period of observation is too short.
- Data should, if possible be correlated with data of nearby stations to check for validity.
- Also data should be checked for possible observation and instrument errors.

Assignment 4

i. Table 1 presents weekly rainfall, temperature, and cup anemometer reading collected at 0900 hours over a period of six (6) weeks in 2013 at DABE Weather Station.

ii. What is rainfall for the period

iii. What is mean weekly temperature

iv. What is mean temperature for the period

v. Calculate the standard deviation and CV for weekly rainfall and temperature

vi. Based on your answers in v, compare and comment on the variability in the weekly rainfall and temperature

vii. What is the highest weekly wind speed (km/h)

viii. Which week is likely to have the highest and lowest Relative Humidity and why

Assignment 4

Table 1 presents weekly rainfall, temperature, and cup anemometer

Week	Rainfall (mm)	Max Temp. (° C)	Min Temp. (° C)	Anemometer Reading (km)	Dry Bulb Temp. (° C)	Wet Bulb Temp. (° C)
1	25.4	29.3	21.9	5917.1	24.3	22.7
2	48.2	30.0	21.6	6446.5	23.8	22.4
3	3.2	29.1	21.5	7033.9	23.3	22.0
4	64.9	30.2	21.5	7462.5	23.9	22.6
5	39.4	30.1	21.3	7894.3	22.7	22.2
6	15.6	29.5	20.8	8412.4	23.5	22.3
7				8848.7		

Missing data point or rainfall at ungauged locations (1/6)

1. Simple Average of surrounding gauges

   Used for:

   i. Missing data estimation

   ii. Ungauged location data estimation

   Requirement/Conditions:

   a. Gauges must be of about equal distance from missing data point

   b. Gauges should have fairly equal data (e.g. rainfall) at known gauges

Missing data point or rainfall at ungauged locations (2/6)

2. Normal Ratio Method for missing data estimation
   - Stations are weighted based on normal annual data (precipitation) ratio's

Used for:

- Missing data estimation

Requirement/Conditions :

- When surrounding gauges differ by more than 10% in measured rainfall

Missing Data Point or Rainfall at Ungauged Locations (3/6)

Normal Ratio Method for missing data

$$P_x = \frac{N_x}{k}\left(\frac{P_1}{N_1} + \frac{P_2}{N_2} + ... + \frac{P_k}{N_k}\right)$$

Where:

N_x = normal long-term annual data (precipitation) at location x (i.e. location of interest)

P_k = event data (precipitation) at gauge k

N_k = normal long-term annual precipitation at gauge k

P_x = estimated event precipitation at location x (i.e. location of interest)

K = n - 1

The event can be a single storm, daily total, monthly total or annual value.

Sample Question

- Table 1 presents the monthly rainfall for June 2012 and long term mean annual rainfall for each of the rain gauge locations.
- Using the simple arithmetic average calculate mean monthly rainfall for June 2012 for the watershed
- Write an expression for the estimation of missing data using Normal Ratio Method
- Estimate the missing rainfall data for June 2012 at location P_6 using the expression and the data in Table …

Sample Question

Table 1 presents the monthly rainfall for June 2012

Rain gauge	Monthly rainfall for June 2012 (mm)	Long term annual rainfall (mm)
P_1	115	1300
P_2	156	1550
P_3	140	1350
P_4	150	1448
P_5	130	1288
P_6	X	1290
P_7	152	1524

Solution

Rain gauge	Monthly rainfall for June 2012 (mm) (Pi)	Long term annual rainfall (mm) (Ni)	P_k/N_k
P_1	115	1300	0.08846
P_2	156	1550	0.10065
P_3	140	1350	0.10370
P_4	150	1448	0.10359
P_5	130	1288	0.10093
P_6	X	1290	
P_7	152	1524	0.09974
Total			0.59707
Mean	**140.5**	Px	**128.37**

Sample Question

Estimate the missing rainfall data at location P_3

Rain gauge	Monthly rainfall for May 2014 (mm)	Long term annual rainfall (mm)
P_1	110	1200
P_2	89	900
P_3	X	1350
P_4	120	1458
P_5	95	1188
P_6	116	1145
P_7	122	1500

Missing data point or rainfall at ungauged locations (4/6)

3. Reciprocal (Inverse weighted) Distance Method

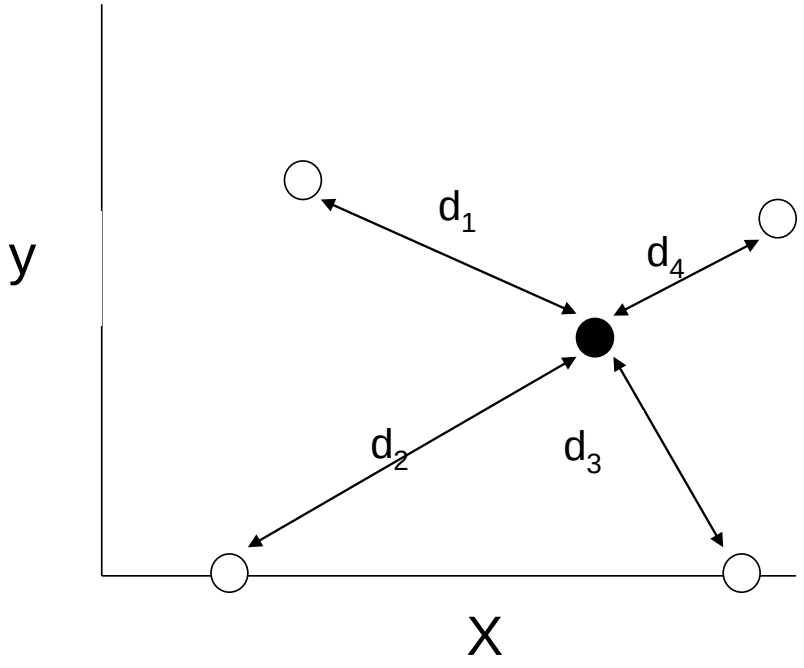
Principle: Surrounding gauges are weighted based on squares of reciprocals of distance away from the point of interest

Used for:

i. missing data estimation
ii. Un-gauged location data estimation

Missing data points or rainfall at ungauged locations (5/6)

Reciprocal Distance Method (2/3)



1. Distance away from the point of interest

$$d_i = \sqrt{x^2 + y^2}$$

2. X and y are the horizontal and vertical distances from the un-gauged (missing data) location where precipitation is being estimated

Missing data points or rainfall at ungauged locations (6/6)

Reciprocal Distance Method (3/3)

$$W_i = \frac{1}{d_i^2}$$

$$P_x = \frac{\sum_{i=1}^{n} P_i W_i}{\sum_{i=1}^{n} W_i}$$

Sample Question

Rain gauge	Monthly rainfall for June 2012 (mm)	Long term annual rainfall (mm)	Distance from P6 (km)
P_1	115	1300	7
P_2	156	1550	15
P_3	140	1350	20
P_4	150	1448	11
P_5	130	1288	4
P_6	X	1290	0
P_7	152	1524	14

Solution

Rain gauge	Monthly rainfall for June 2012 (mm) (Pi)	Distance from P6 (km) (d)	d^2	$W_{i=1/d^2}$	PiW_i
P1	115	7	49	0.02041	2.346939
P2	156	15	225	0.00444	0.693333
P3	140	20	400	0.0025	0.35
P4	150	11	121	0.00826	1.239669
P5	130	4	16	0.0625	8.125
P6	X	0	0		
P7	152	14	196	0.0051	0.77551
Total				0.103219	13.52899
Estimate	**140.5**			Px	**131.07**